DISCUSSION DRAFT · OCTOBER 2026

# OECMs in Tunisia: where to start

A site-by-site screening of other effective area-based conservation measures, and recommendations to reach Target 3 of the Kunming-Montreal Global Biodiversity Framework (30×30).

Based on 2,406 candidate sites mapped by the OECM Harvester (polygon layers), the GBF Headline Indicator 3.1 analysis of Tunisia (WDPA/WD-OECM, October 2026) and the World Database of Key Biodiversity Areas (September 2026).

**9 + 2**

priority-1 complexes + marine zones to report to the WDPA

**34%**

of the sea with priority 1 and 2 (today 1.07%)

**9.0%**

of the land with priority 1 and 2 (today 7.93%)

**42 of 70**

KBAs with a gap that priority 1 and 2 reach

Screening for discussion with the Tunisian administrations and partners. Priorities come from a provisional, transparent rubric; recognising an OECM needs the consent of its governing authority and a full assessment against the CBD criteria.

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### COMPANION MATERIALS

1. **Interactive dashboard**
   north_africa_dashboard/wwf_site_oecm, tab OECM opportunities: every site on a map, with its full profile.
2. **Data**
   tunisia/06_oecm/outputs: site complexes (CSV, GeoPackage), KBAs, scenarios, every candidate.
3. **Method and judgements**
   tunisia/06_oecm/scripts: the analysis and oecm_measures.yaml, which holds every expert judgement.
4. **This report**
   tunisia/06_oecm/report: rebuilt from the data by DesignCraft whenever the analysis changes.

Orange squares: priority-1 sites, two pages each (description, then assessment against the OECM criteria). Blue squares: marine zones that are protected but missing from the WDPA (reporting gaps).

ABOUT THIS REPORT

# Why this report, and how to read it

Tunisia has committed, with every other Party to the Convention on Biological Diversity, to conserve 30% of its land and of its sea by 2030. Protected areas alone will not get it there in time. This report looks at the other half of Target 3: places that are already managed in ways that conserve nature, even though conservation is not their main purpose.

Dams and their shores, fishing grounds closed to trawlers, hunting reserves, oases protected by agricultural law, sacred groves, archaeological sites and military land all restrict, in one way or another, the activities that damage biodiversity. Some of them could be recognised as **other effective area-based conservation measures (OECMs)** and counted towards Target 3. Others cannot, because their protection is temporary, too narrow or not linked to biodiversity. Telling the two apart, site by site, is what this report sets out to do.

The analysis starts from the **OECM Harvester**, a tool that maps, category by category, the sites that could be OECMs, from open data and from national legal texts. For Tunisia it mapped 2,406 sites with a boundary in eight categories. Each of them was measured against the protected areas already counted in Headline Indicator 3.1, the Key Biodiversity Areas (KBAs), land cover and the four criteria that CBD Decision 14/8 sets for OECMs. The result is a ranked list, a set of scenarios and a written case for every priority site.

## Who it is for

The report is written for the administrations that govern the candidate sites and for their partners: the ministry in charge of the environment as CBD focal point, the forest, fisheries, water and heritage administrations, the agency for coastal protection, the Ministry of Defence, regional agriculture commissariats, oasis and local associations, research institutes and conservation NGOs. It is a basis for discussion: none of the sites can become an OECM without the agreement of the authority that governs it.

## How it is organised

- **Key messages and recommendations** (pages 4–5) summarise what to do and who could lead.
- **Context** explains Tunisia's conservation gap, what OECMs are and what makes them credible.
- **Method and results** describe the screening, what OECMs could add under five scenarios, and where the priority sites are.
- **Priority sites** gives two pages to each priority-1 complex: a narrative of the site, then its assessment against the OECM criteria with risks and next steps.
- **The sea, priority 2, KBAs and categories** look at the larger picture; the **roadmap** proposes a national process.

## Reading a site assessment

Each priority site carries a status for each of the four OECM criteria. **Met**: the evidence in hand meets the criterion. **Likely**: the legal basis and the data point that way, to be confirmed. **Partly**: part of the site meets it (for example part of the area is already a protected area). **Uncertain**: evidence is missing. **Unlikely**: as it stands, the criterion is not met.

All figures are computed from open data (WDPA/WD-OECM October 2026, World Database of KBAs September 2026, ESA WorldCover 2021, OECM Harvester) in the equal-area projection used by indicator 3.1. Areas are net of the protected areas already counted, and totals count overlaps once.

### THIS REPORT IN NUMBERS

**2,406** candidate sites with a mapped boundary

**8** categories of measure screened

**72** Key Biodiversity Areas tested

**9 + 2** priority-1 complexes + marine zones to report

**5** scenarios, from today to every candidate

**4** OECM criteria, assessed site by site

SUMMARY

# Key messages

1

**Report what is already protected.**

Two marine park zones (Zembra–Zembretta zone C and the Kneiss islands) are protected but missing from the WDPA records counted in indicator 3.1. Reporting them adds 117 km² of sea with no new measure.

2

**Start with 9 priority-1 complexes.**

Seven dam reservoirs whose shores are Key Biodiversity Areas (KBAs) for waterbirds or freshwater species, the no-fishing zone around La Galite and the archaeological site of Henchir Bourgou. Each combines a permanent legal restriction, natural habitat and a KBA: the best pilots for a national OECM process.

3

**At sea, the trawling bans decide 30%.**

The permanent bottom-trawling bans (Gulf of Gabès and Gulf of Tunis under 50 m, the 3-mile coastal band) and the deep-sea GFCM restricted area would take sea coverage from 1.07% to about 34%, if Tunisia can show that they protect seagrass and nurseries and that illegal trawling is controlled.

4

**On land, these categories will not close the gap.**

Priority 1 and 2 complexes add 1,662 km² (+1.07 points) against a gap of 34,340 km². New protected areas and other OECM types (managed state forests, rangeland reserves, community areas) are needed.

5

**OECMs can close some KBA gaps.**

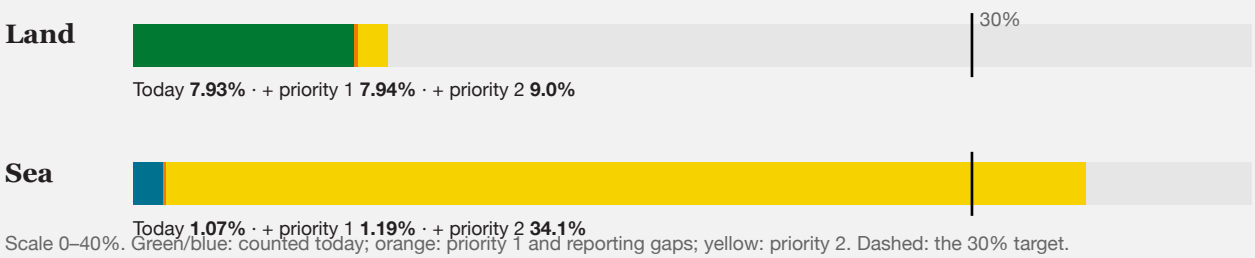
Priority 1 and 2 complexes reach 42 of the 70 KBAs that are not fully covered and close at least half of the gap in 13. 23 KBAs are reached by no candidate at all.

6

**Small sites matter too, after checks.**

Wooded shrines in the Mogods and Kroumirie, traditional oases and archaeological sites score well but need field checks, community consent and biodiversity evidence before any recognition.

## WHERE TUNISIA STANDS, AND WHERE OECMS COULD TAKE IT




Scale 0–40%. Green/blue: counted today; orange: priority 1 and reporting gaps; yellow: priority 2. Dashed: the 30% target.

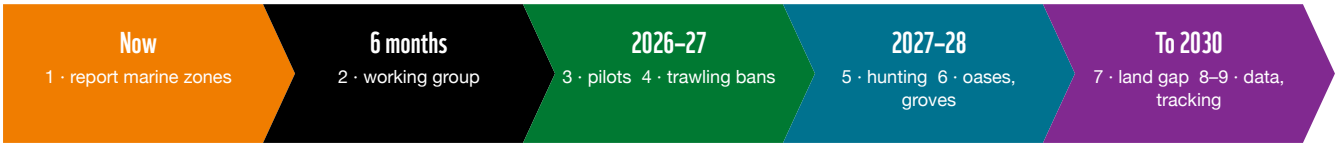
WHAT TO DO

# Recommendations at a glance

Nine actions, in the order they can be taken. The first two cost little and can start now; the others build the national process that Target 3 needs.

| # | Recommendation | Lead and partners | When |
|---|---|---|---|
| 1 | **Report the two marine zones already protected** (Zembra–Zembretta zone C, Kneiss islands) to the WDPA through UNEP-WCMC. | Ministry in charge of the environment, APAL | Now (2026) |
| 2 | **Set up a national OECM working group** and adopt a recognition procedure based on CBD Decision 14/8 and the IUCN-WCPA site-level tool. | Ministry in charge of the environment (CBD focal point) with DGF, DGPA, DGBGTH, APAL, INP, Ministry of Defence; NGOs | Next 6 months |
| 3 | **Assess the 9 priority-1 complexes** as pilots: confirm boundaries and legal status, compile waterbird and species data, obtain consent. | DGBGTH and DGF (dams), DGPA (La Galite), INP (Henchir Bourgou); AAO/BirdLife, universities | 2026–2027 |
| 4 | **Decide on the permanent trawling bans**: compile enforcement and seagrass evidence; if conclusive, report them as fisheries OECMs. | DGPA, INSTM, GFCM; National Guard (maritime) | 2026–2027 |
| 5 | **Make stable hunting reserves permanent** and narrow whole-imada reserves to their natural habitat before any OECM discussion. | DGF and regional forest districts (CRDA) | 2027 |
| 6 | **Document oases and sacred groves** with the communities: date varieties and agro-biodiversity, customary rules, grove checks on imagery. | CRDA and oasis water-user groups (GDA), GIAHS team (Gafsa), religious affairs, local associations | 2027–2028 |
| 7 | **Plan for the land gap**: new protected areas and other OECM types (managed state forests, rangelands, community areas), starting with the 23 KBAs no candidate reaches. | Ministry in charge of the environment, DGF, CRDA, communities | 2027–2030 |
| 8 | **Fix the data**: official boundaries of hunting reserves, Water Code belts and heritage perimeters; map sites known only as points. | DGF, DGBGTH, INP, national mapping agency | Continuous |
| 9 | **Track progress in indicator 3.1** each year with the national tool, and report recognised OECMs to the WD-OECM. | CBD focal point with UNEP-WCMC | Yearly |

## TIMELINE




### Quick wins

1 · 2

Low cost, start now: report what is already protected and set up the process.

### Building blocks

3 · 4 · 5 · 6

Pilots and decisions with each authority: dams, trawling bans, hunting reserves, oases and groves.

### Long term

7 · 8 · 9

Close the land and KBA gaps with new protected areas, better data and yearly tracking.

# Tunisia's nature and the 30×30 gap

Tunisia packs a remarkable range of ecosystems into a small country: from the cork-oak and zeen-oak forests of Kroumirie and the Mogods in the north-west, across the Dorsale and the steppes of the centre, to the chotts and the edge of the Sahara in the south; and at sea, from the rocky coasts and islands of the north to the shallow, seagrass-covered shelf of the Gulf of Gabès.

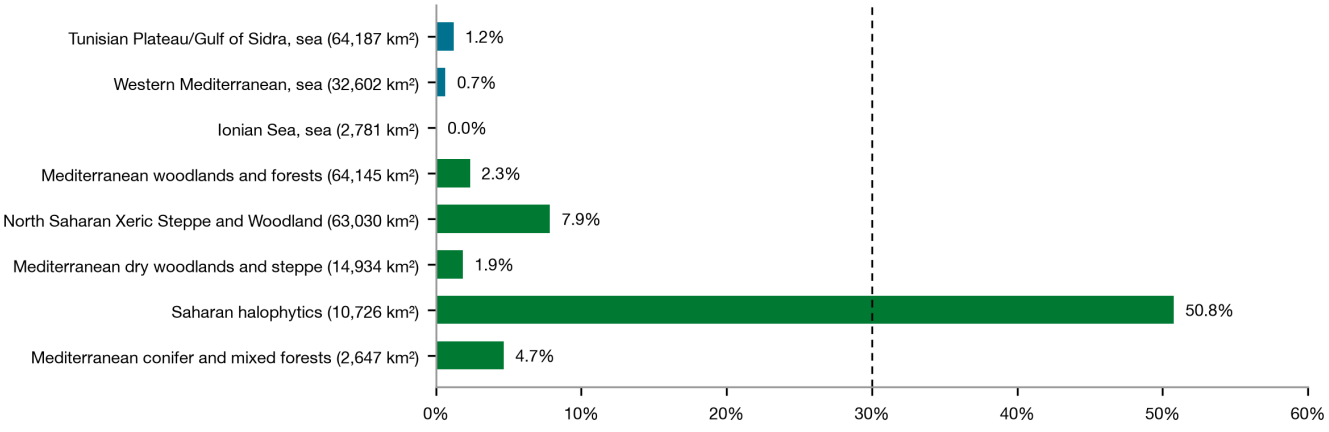
Its protected-area network has grown around this diversity. The World Database on Protected Areas lists 149 records for Tunisia, of which 139 are counted in indicator 3.1 (proposed sites and UNESCO biosphere reserves are not). Most are wetlands: 45 wetland zones of national importance and 42 Ramsar sites, alongside 25 nature reserves and 17 national parks. Almost all (137 of 139) are governed by national ministries or agencies, and only 1.4% of the covered land has a management-effectiveness evaluation less than five years old.

Together these sites cover **7.93% of the land** and **1.07% of the sea**. The government's own figure, in its 7th National Report, is 7.86% land, 1.1% sea; its national target is to add 4% to the protected-area network (national target A3). Reaching 30% would need another **34,340 km² on land** and **28,926 km² at sea**.

Coverage is also uneven. Wetlands are well represented (47% of lakes and wetlands lie in counted sites, helped by the large protected chotts of the south), but only 10% of river length is covered, and the ecoregions where most Tunisians live and farm are barely protected: 2.4% of the Mediterranean woodlands and forests, 1.9% of the dry woodlands and steppe, 4.7% of the conifer and mixed forests. At sea, the Tunisian Plateau, which holds the Gulf of Gabès, is 1.2% covered.

Tunisia has 72 Key Biodiversity Areas: 20 of global significance, 13 regional and 39 whose level is still to be assessed. 45 of them hold at least one threatened trigger species. Counted protected areas cover 41% of an average KBA, and 6,174 km² of KBA lie outside any counted site.

## Coverage of each ecoregion today




Share of each terrestrial (green) and marine (blue) ecoregion inside the protected areas counted in indicator 3.1. Dashed line: 30%. RESOLVE Ecoregions 2017 and Marine Ecoregions of the World, clipped to Tunisia.

# What OECMs are, and why they fit Tunisia

**7.93%**
of the land counted today; 34,340 km² short of 30%

**1.07%**
of the sea counted today; 28,926 km² short of 30%

**6,174 km²**
of KBA not covered, out of 14,808 km²

**2,406**
candidate sites with a mapped boundary, in 8 categories

Target 3 of the Kunming-Montreal Global Biodiversity Framework asks countries to conserve at least 30% of land and sea by 2030 through protected areas **and other effective area-based conservation measures (OECMs)**. Headline Indicator 3.1 counts both. Tunisia counts 7.93% of its land and 1.07% of its sea today, and reports no OECM.

An OECM is not a protected area: it is a place managed for another purpose (water supply, fisheries, defence, heritage, worship, farming) that nevertheless delivers lasting conservation of biodiversity. Recognising such places credits what is already conserved, brings new partners into conservation and can reach places that protected areas miss. It does not create new restrictions by itself, but it does require the consent of the authority that governs the site and evidence of outcomes.

This report screens every candidate mapped so far by the OECM Harvester for Tunisia against the four OECM criteria of CBD Decision 14/8, measures what each would add to indicator 3.1 and to the Key Biodiversity Areas, and proposes where to start.

| | Protected area | OECM |
|---|---|---|
| **Main purpose** | Conservation | Another purpose (water, fishing, heritage…) |
| **Outcome required** | Conservation | Lasting conservation, whatever the purpose |
| **Counted in 3.1** | Yes | Yes, once reported |
| **Tunisian example** | Ichkeul National Park | A dam whose shores are a KBA |

## The four OECM criteria (CBD Decision 14/8, Annex III)

### A
**Not a protected area**
The area is not recognised or reported as a protected area. Places already inside counted protected areas add nothing.

### B
**Governed and managed**
A geographically defined area, a legitimate governing authority (state, community, private) and management that sustains the outcomes: a law, a regulation, a customary rule.

### C
**Sustained in-situ biodiversity outcomes**
Effective, long-term conservation of biodiversity in place: species and habitats, ideally of KBA importance, shown by information and monitoring. Temporary or seasonal measures rarely qualify.

### D
**Associated values**
Ecosystem functions and services, and cultural, spiritual or socio-economic values, are recognised and respected.

**How this screening uses them.** Criterion A is measured exactly (net area outside the protected areas counted in 3.1). B and C are judged from the legal basis, the permanence and strength of the measure, KBA overlap, threatened trigger species and land cover. D is described. Each priority site below carries a status for each criterion: met, likely, partly, uncertain or unlikely.

# What makes an OECM credible

OECMs can strengthen Target 3, or weaken it if they become a way to count places that do not conserve anything. The CBD guidance and IUCN's work on OECMs point to five conditions, which this report applies site by site.

## 1. Tunisia already has the measures

The country's laws contain many area-based rules that restrict damaging activities without being conservation law: the **Water Code** (art. 123) bans polluting activities in belts around drinking-water reservoirs; the **fishing order of 28 September 1995** closes zones to all fishing or to trawling; the yearly **hunting order** (art. 12) lists game reserves where hunting is banned; the **Heritage Code** (Law 94-35) protects archaeological sites; **Law 83-87** makes oases agricultural safeguard zones; and a **2013 order** closes the southern border zone. The OECM Harvester rebuilt many of these zones from their legal texts.

## 2. Biodiversity outcomes, not intentions

An OECM must deliver lasting conservation of biodiversity in place, whatever its purpose. The strongest cases are sites where the measure protects habitat that matters: a KBA, threatened species, a natural habitat that is rare in its ecoregion. The weakest are measures that remove one pressure among many (a single fishing gear), apply for part of the year, or cover villages and farmland.

## 3. Long-term protection

Permanence in law, or in practice over many years, is required. Seasonal fishing closures and rotating hunting reserves fail this test; hunting reserves listed in every recent order, and reservoirs owned by the state, pass it more easily.

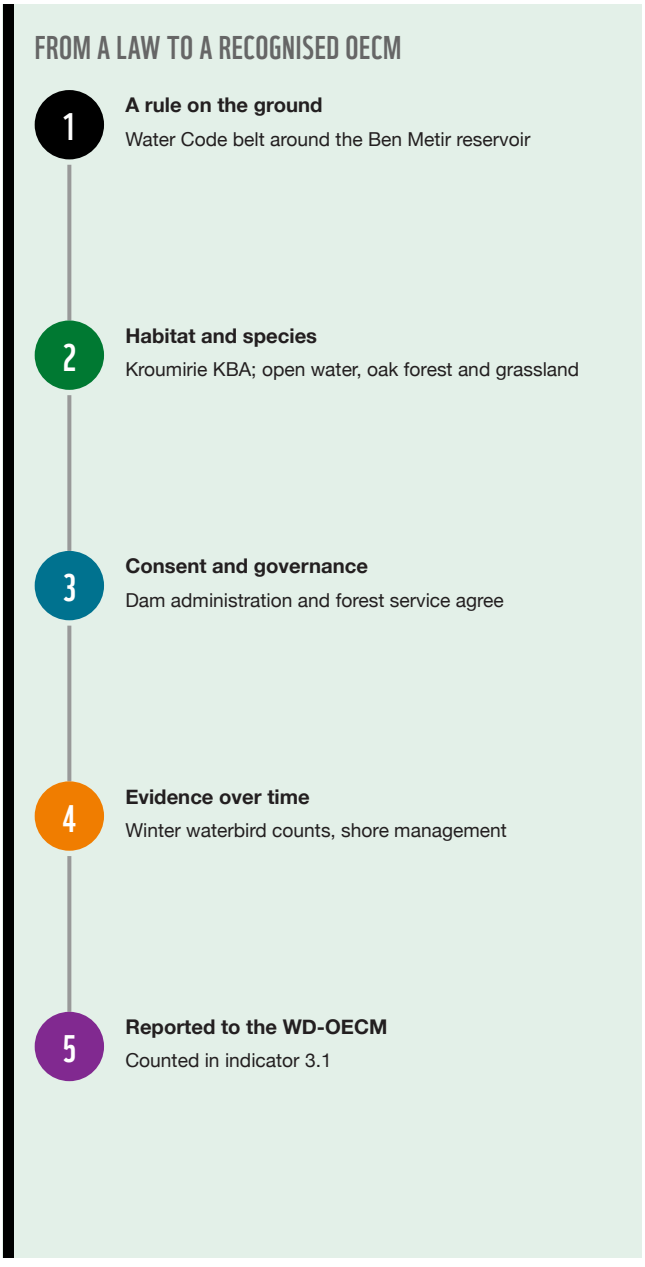
## 4. Governance and consent

Every candidate has a governing authority: a ministry, a public agency, a community, a religious body, the army. Recognition needs its agreement and a clear role in monitoring. It must not reduce the rights of the people who use the site, from oasis farmers to the custodians of a shrine; their knowledge is often what keeps the site intact.

## 5. Evidence and reporting

Recognised OECMs are reported to the World Database on OECMs and counted in indicator 3.1. Each needs a boundary, a description of the measure and of its biodiversity values, and a plan to check outcomes over time: waterbird counts on reservoirs, seagrass and fish surveys at sea, vegetation surveys on heritage sites.

**In this report**, conditions 2 and 3 drive the score (biodiversity and permanence), condition 4 shapes the next steps (who to involve), and condition 5 explains why boundaries and data are part of the recommendations.



# How the sites were screened

**1**

### Map the candidates

The OECM Harvester gathers, by category, the sites that could be OECMs from open data and legal texts: fisheries closures, military areas, hunting reserves, dam reservoirs and Water Code protection zones, traditional oases, sacred groves and cemeteries, archaeological and cultural sites. Only sites with a mapped boundary are used (2,406 polygons).

**2**

### Remove what is already counted

Each site is cut to the realm its measure applies to (sea for fisheries, land for the rest), then the protected areas and OECMs already counted in indicator 3.1 are removed, in the equal-area projection used by the indicator.

**3**

### Measure the evidence

For the remaining (net) area: overlap with each KBA and with its uncovered part; threatened trigger species; land cover (ESA WorldCover 2021, 10 m); ecoregion and how well it is covered; distance to the nearest protected area; governorate.

**4**

### Group records of the same place

A dam, its Water Code belts and the hunting reserve on it; fragments of one oasis: 2,406 records form 2,094 site complexes, assessed together.

**5**

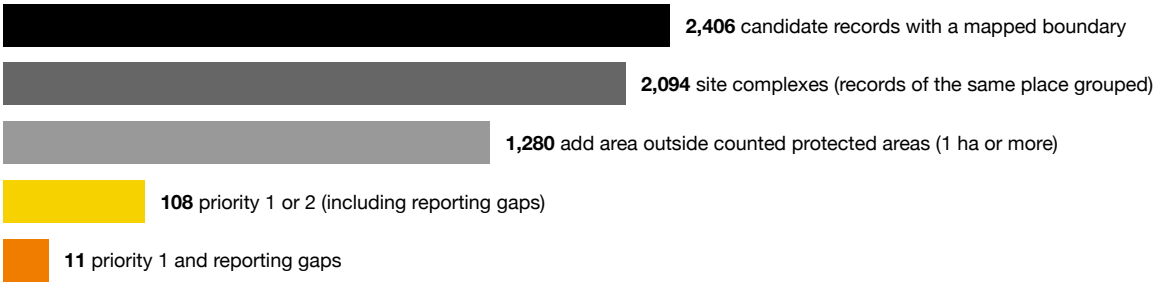
### Score and rank

Score = 0.45 × measure (strength, permanence, mapping confidence) + 0.40 × biodiversity (KBA overlap weighted by habitat, natural habitat, ecoregion representation, connectivity) + 0.15 × size, plus a bonus when several measures stack.

**6**

### Set priorities and argue

Priority 1 from 0.60 (at least 10 ha, mapped with medium confidence), priority 2 from 0.48 (at least 5 ha), else priority 3. Temporary measures are priority 3; trawling-only bans, whole imadas and cemeteries are priority 2 at most; the border buffer zone priority 3. Each complex gets a written case against criteria A–D, risks, evidence to collect and next steps.

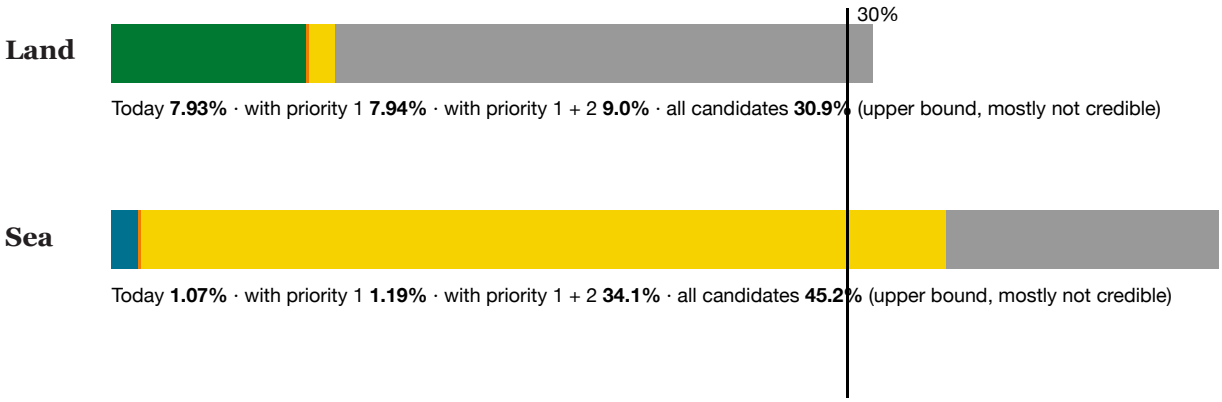



## Limits

Candidate outlines come from open data of uneven quality (OpenStreetMap, reservoir extents, legal zones rebuilt from text, tree-cover detections). KBA overlap shows potential, not the presence of trigger species. Land cover is a 2021 snapshot. Strength, permanence and confidence are expert judgements per measure type. The results open a discussion; they are not an assessment.

# What OECMs could add

Recognising the priority-1 complexes and reporting the two marine zones would add little area, but the right kind. Priority 2 decides the sea target. On land, all candidates together would not reach 30% credibly.




Green/blue: counted today. Orange: priority 1 and reporting gaps. Yellow: priority 2. Grey: priority 3. Overlaps between complexes are counted once. Scale 0–50%.

| Category | Records | Complexes | Priority 1* | Priority 2 | Priority 3 | Covered / <1 ha |
|---|---|---|---|---|---|---|
| Dams & reservoirs | 338 | 320 | 7 | 25 | 284 | 4 |
| Traditional oases | 318 | 64 | 0 | 23 | 41 | 0 |
| Hunting reserves | 142 | 142 | 1 | 19 | 122 | 0 |
| Sacred natural sites | 1,066 | 1,066 | 0 | 14 | 630 | 422 |
| Archaeological sites | 321 | 321 | 1 | 12 | 62 | 246 |
| Fisheries closures | 19 | 19 | 3 | 5 | 4 | 7 |
| Military areas | 18 | 18 | 0 | 4 | 13 | 1 |
| Cultural & historic sites | 184 | 184 | 0 | 1 | 25 | 158 |

* including reporting gaps. Complexes are counted in every category they contain.

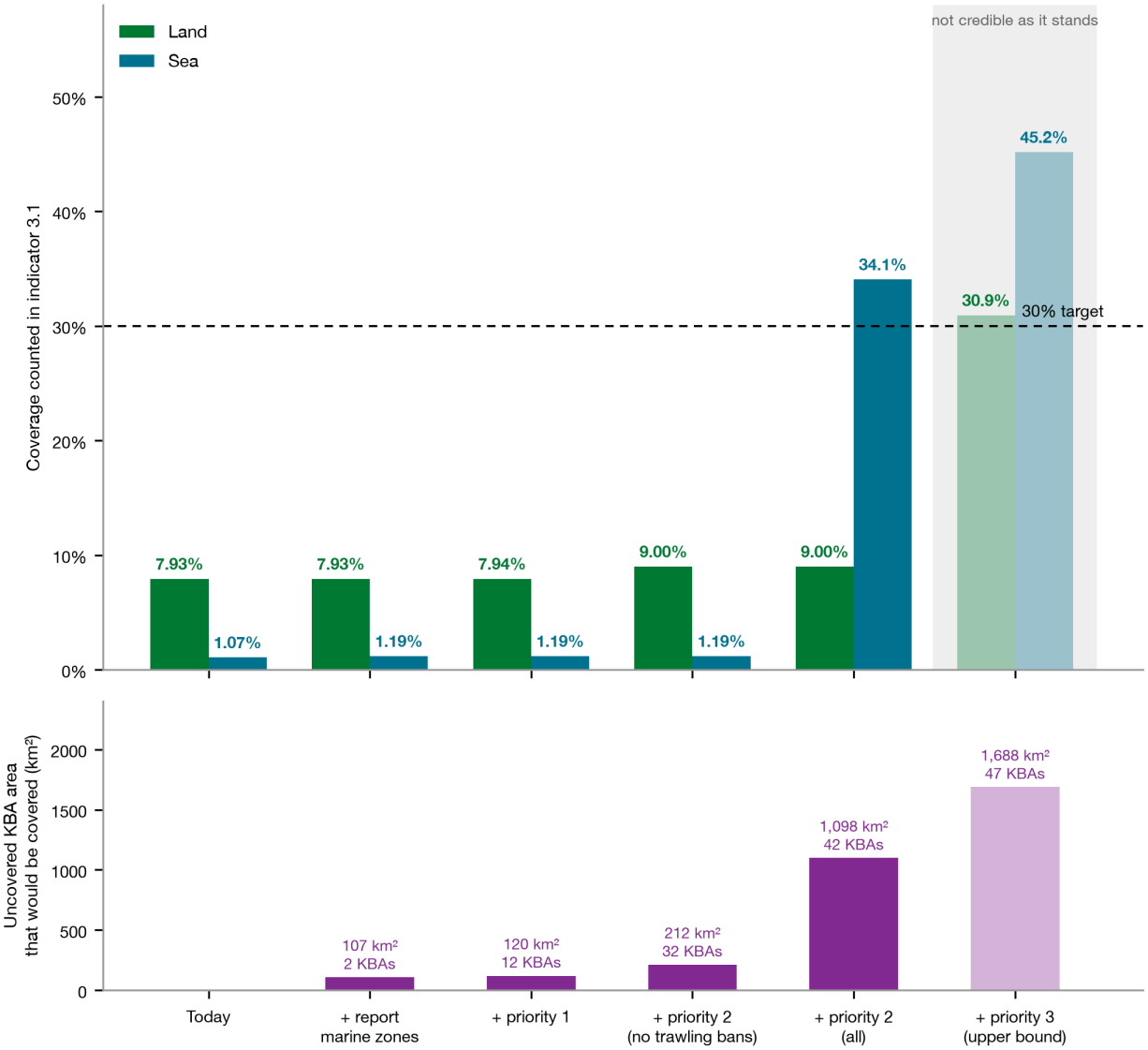
**Dams and reservoirs** provide most priority-1 sites: seven reservoirs whose shores are KBAs and where the Water Code and hunting bans already stack. **Fisheries** closures dominate in area. **Oases**, **hunting reserves** and **sacred groves** supply most priority-2 complexes: valuable, but each needs evidence or narrowing first.

## PRIORITY 1 + 2 COMPLEXES: HOW MANY, AND HOW MUCH AREA, BY CATEGORY



# Five scenarios: how far OECMs could take Tunisia

What would Tunisia count if it recognised its OECM candidates step by step, starting with the strongest? Each scenario adds a set of site complexes to the protected areas counted today; overlaps are counted once.




Top: share of land (green) and sea (blue) counted in indicator 3.1 under each scenario; dashed line: 30%. Bottom: area of Key Biodiversity Areas, not covered today, that each scenario would cover, and how many KBAs it reaches. The last scenario is an upper bound: it includes temporary, rotating and security measures that would not qualify as they stand.

| Scenario (cumulative) | Complexes | Land | Sea | + land km² | + sea km² | KBA gap km² | KBAs |
|---|---|---|---|---|---|---|---|
| **Today** | 0 | 7.93% | 1.07% | 0 | 0 | 0 | 0 |
| **Report marine zones** | 2 | 7.93% | 1.19% | 0 | 117 | 107 | 2 |
| **Priority 1** | 11 | 7.94% | 1.19% | 13 | 122 | 120 | 12 |
| **Priority 2 without trawling bans** | 104 | 9.00% | 1.19% | 1,662 | 122 | 212 | 32 |
| **Priority 2** | 108 | 9.00% | 34.06% | 1,662 | 32,984 | 1,098 | 42 |
| **Priority 3 (upper bound)** | 1280 | 30.93% | 45.17% | 35,795 | 44,093 | 1,688 | 47 |

# Reading the scenarios

## Scenario 1: report what is already protected

Two zones of marine protected areas, recorded by ProtectedSeas but missing from the WDPA, would add **117 km²** of sea (sea coverage 1.19%) and cover 107 km² of the Zembra and Kneiss KBAs. No new measure is needed: only a submission to UNEP-WCMC. It is the cheapest gain in this report.

## Scenario 2: add the priority-1 complexes

The nine priority-1 complexes add 13 km² of land and 4.6 km² of sea. That moves the indicator by hundredths of a point, but it reaches 12 KBAs and closes more than half of the uncovered part of 2, including the White-headed Duck reservoirs of Masri and Bezikh. Their value is quality and precedent: they are the strongest cases, managed by the state, with a permanent legal basis, and they would test the national recognition procedure.

## Scenario 3a: add priority 2 on land

Adding the priority-2 oases, reservoirs, hunting reserves, sacred groves, heritage and military sites (but not the trawling bans) brings land coverage to **9.00%** (+1,662 km²) and reaches 32 KBAs. Each of these complexes needs something first: a field check, a narrower boundary, a permanent status or the consent of an authority that is not used to conservation. Even if all of them were recognised, land coverage would stay far below 30%.

## Scenario 3: add the permanent trawling bans

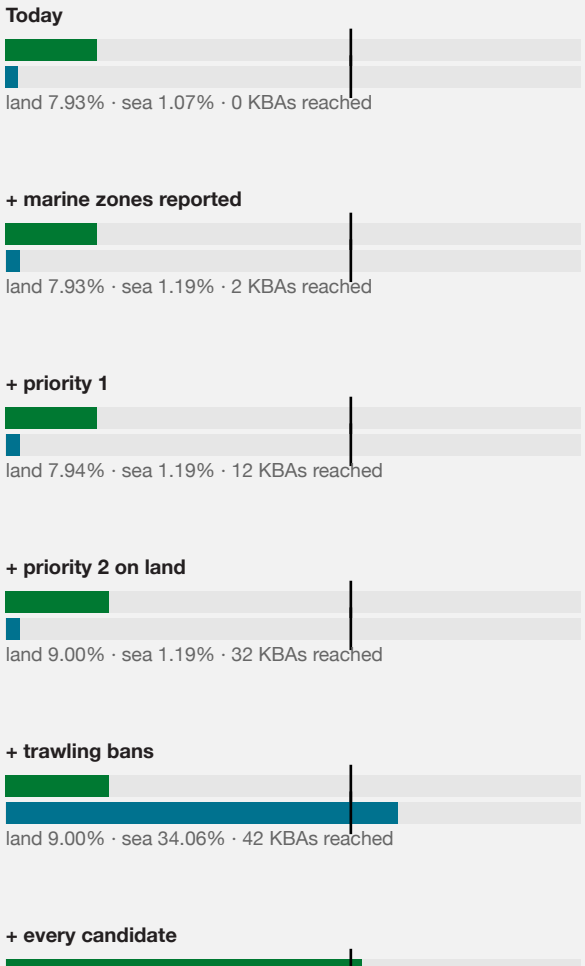
The permanent bans on bottom trawling (Gulf of Gabès and Gulf of Tunis under 50 m, the 3-mile coastal band) and the deep-sea GFCM restricted area change the picture at sea: coverage jumps to **34.1%**, above 30%, and the KBA gap covered rises to 1,098 km² in 42 KBAs. Whether this is credible depends on evidence that Tunisia does not yet publish: how well the bans are enforced, and the state of the seagrass meadows and nurseries they are meant to protect. This is the single most important decision for Target 3 at sea.

## Scenario 4: every candidate (an upper bound)

Counting everything the Harvester mapped would reach 30.9% of the land and 45.2% of the sea. Most of that land is the southern border zone (a security measure renewed yearly) and whole hunting imadas that include villages and farmland. This scenario is not a target: it shows that the land gap cannot be closed by relabelling existing measures, and that new protected areas, and OECM types not yet mapped (managed state forests, rangeland reserves, community-conserved areas), are needed.

Scenarios are cumulative. Areas are net of the protected areas counted today and of each other. The KBA figures count only KBA area that no counted protected area covers today.

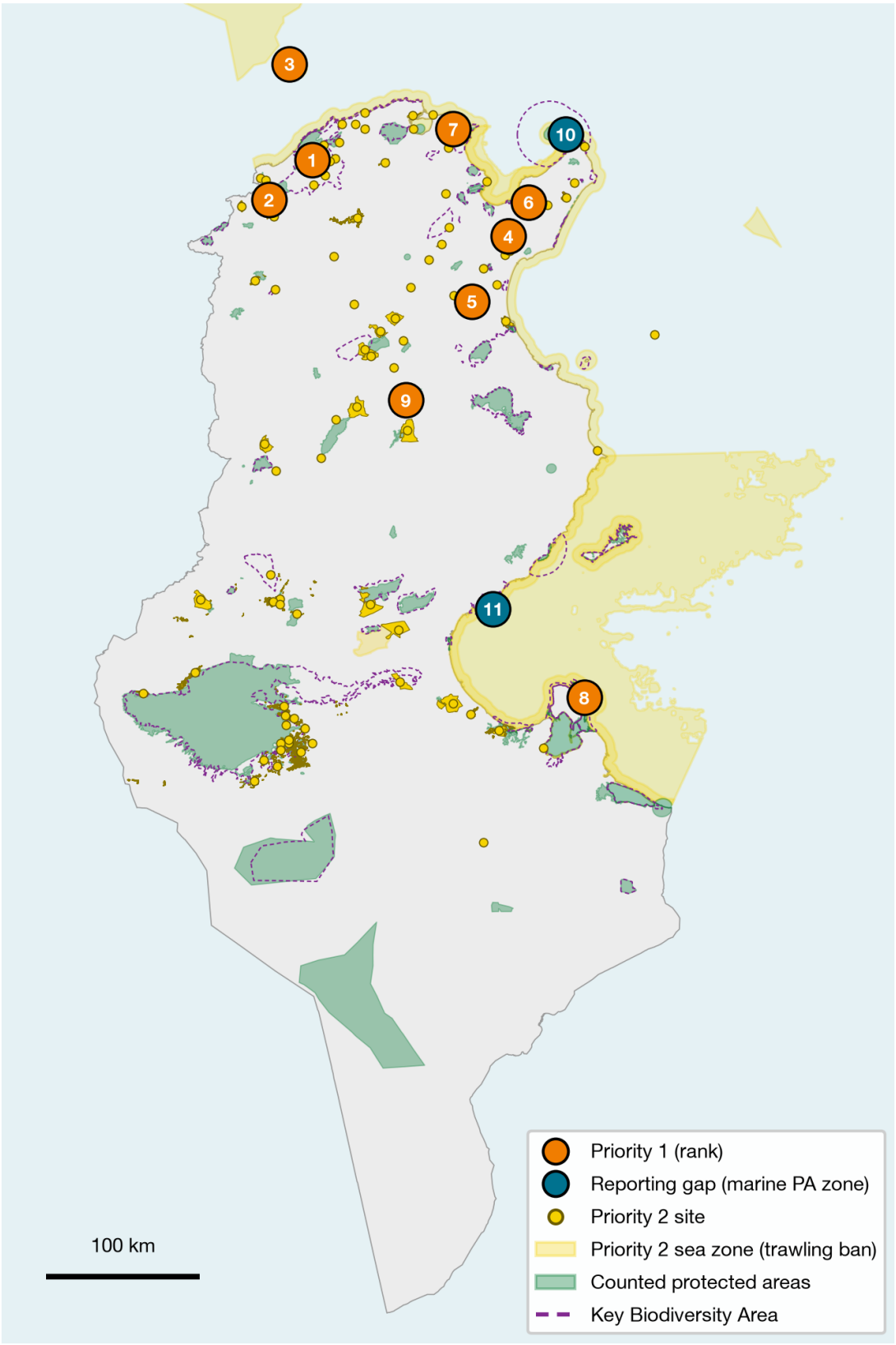
### THE STAIRCASE




### THE KEY DECISION

The permanent trawling bans alone add 32.9 points at sea: whether they qualify decides Target 3 at sea.

# Where the priority sites are




Numbered circles: priority-1 complexes (orange) and marine zones to report to the WDPA (blue), by rank (see the list on the next page). Yellow: priority-2 sites and, at sea, the permanent trawling bans. Green: protected areas counted in indicator 3.1. Dashed: Key Biodiversity Areas. Projection UTM 32N.

# Priority 1: the pilots

9 complexes meet every condition for priority 1: a permanent legal restriction, mostly natural habitat, a KBA, at least 10 ha outside counted protected areas and a reliable outline. Two marine zones are protected areas missing from the WDPA.

Seven of the nine are dam reservoirs. Tunisia has few natural freshwater wetlands outside the chotts, and several of its reservoirs have become wintering and breeding sites for waterbirds, among them the White-headed Duck (Endangered), which is the trigger species of the Masri, Bezikh, El Ogla and Khairat KBAs.

administration, partly protected by the Water Code and sometimes listed as hunting reserves: the ingredients of an OECM are already there. The two other priority-1 sites are the strict no-fishing zone around La Galite and the archaeological site of Henchir Bourgou on Djerba. Two pages per site follow: a description, then the assessment.

| # | Complex | Governorate | Measures | Net ha | KBA | Threatened trigger species |
|---|---|---|---|---|---|---|
| 1 | **Sidi el Barrak dam** | Béja | Hunting reserve on a named site (every season); Water Code protection zone; Dam reservoir or hill lake | 618 | Maden River, Oued El Zouara | Anodonta lucasi (CR); Calopteryx exul (EN); Convolvulus durandoi (CR)… |
| 2 | **Ben Metir dam** | Jendouba | Water Code protection zone; Dam reservoir or hill lake | 387 | Kroumirie | Pilularia minuta (EN) |
| 3 | **No-fishing zone around La Galite and Galiton (1.5 mile)** | Bizerte | No-fishing zone | 453 | Archipel de la Galite | – |
| 4 | **El Masri dam** | Nabeul | Dam reservoir or hill lake | 77 | Barrage Masri | Oxyura leucocephala (EN) |
| 5 | **El Ougla dam** | Sousse | Dam reservoir or hill lake | 76 | Barrage Khairat | Marmaronetta angustirostris (VU); Oxyura leucocephala (EN) |
| 6 | **Bezirk dam** | Nabeul | Dam reservoir or hill lake | 68 | Barrage Bezikh | Oxyura leucocephala (EN) |
| 7 | **Dam reservoir or hill lake, Bizerte (GDW 28423)** | Bizerte | Dam reservoir or hill lake | 11 | Jbel Nadhour et Lagune de Ghar El Melh | Acanthodactylus blanci (EN) |
| 8 | **Henchir Bourgou** | Médenine | Archaeological site | 21 | Île de Djerba | – |
| 9 | **El Habib dam** | Kairouan | Dam reservoir or hill lake; Water Code protection zone | 50 | Barrage El Houareb | Aythya nyroca (VU); Marmaronetta angustirostris (VU); Oxyura leucocephala (EN) |
| 10 | **Parc National de Zembra et Zembretta - zone C, Transition and sustainable development area** | Nabeul | Marine protected area zone not counted | 9,134 | Archipel de Zembra | – |
| 11 | **Iles Kneiss** | Sfax | Marine protected area zone not counted | 2,589 | Îles Kneiss | – |

Net ha: area outside the protected areas counted in indicator 3.1. Rows in blue: marine park zones to report to the WDPA (reporting gaps). Threatened: IUCN Red List CR, EN or VU trigger species of the KBA(s) the complex overlaps.

## PRIORITY 1 AT A GLANCE

**7**
dam reservoirs in KBAs, with Water Code belts or hunting bans

**13 km²**
of land and 122 km² of sea, outside counted PAs

**12**
KBAs reached, 2 of them more than half closed

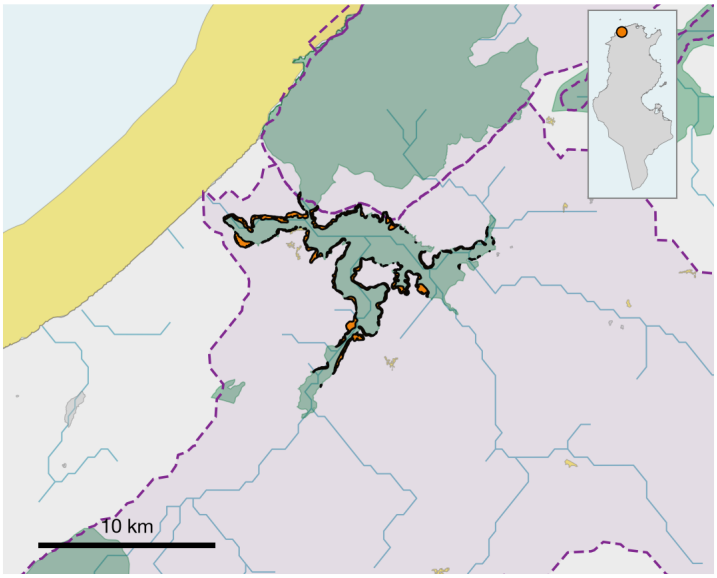
**11**
threatened trigger species in the KBAs they overlap

Algerian Ribbed Newt (VU) · Anodonta lucasi (CR) · Blanc's Fringe-toed Lizard (EN) · Calopteryx exul (EN) · Convolvulus durandoi (CR) · Dwarf Pillwort (EN) · Ferruginous Duck (VU) · Gomphus lucasii (VU) · Marbled Teal (VU) · Serapias stenopetala (CR) · White-headed Duck (EN)

PRIORITY 1 · #1 · BÉJA GOVERNORATE

# Sidi el Barrak dam

Hunting reserve on a named site (every season) · Water Code protection zone · Dam reservoir or hill lake · 4 records of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**618 ha**
outside counted protected areas

**618 ha**
in KBAs (100% of the net area)

**+0.004**
points of % of Tunisia's land

**91%**
natural habitat (WorldCover)

### Why it is a priority

- In Maden River and Oued El Zouara, KBA with 6 threatened trigger species.
- Mostly natural habitat (open water 39%, tree cover 27%, grassland 22% and cropland 6%).
- Several measures on the same ground: hunting reserve on a named site (every season), water code protection zone and dam reservoir or hill lake.
- The restriction is permanent in law.

### The site

Sidi el Barrak dam is a dam reservoir in Béja governorate: built in 2000, holds 275 million m³, drains a catchment of 910 km²; it is used for irrigation first, water supply. Around and on the reservoir, hunting reserve on a named site (every season) and water code protection zone records add their own restrictions, so several measures overlap on the same ground.

Most of the mapped area (88%) already lies inside counted protected areas; only 618 ha around them would add to indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Maden River** (global or regional significance, level to be confirmed, 820.0 km², 11% covered by protected areas today).

The KBA was identified for 8 trigger species, 6 of them threatened on the IUCN Red List: Anodonta lucasi (CR), Sérapias à pétales étroits (Serapias stenopetala, CR), Liseron de durand (Convolvulus durandoi, CR), Calopteryx exul (EN), Algerian Ribbed Newt (Pleurodeles nebulosus, VU) and 1 more.

Satellite land cover (ESA WorldCover 2021) shows the net area as open water 39%, tree cover 27%, grassland 22% and cropland 6%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

### What protects it today

Hunting reserve on a named site (every season); Water Code protection zone; Dam reservoir or hill lake. Legal basis: Code des eaux, art. 123-1; Code des eaux, art. 123-2; hunting order 2025/2026, art. 12. Authorities: Direction Générale des Forêts (DGF, Ministry of Agriculture) and the regional forest district (CRDA); DGBGTH (Ministry of Agriculture, dams) and SONEDE (drinking water); CRDA; DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA). The full case is on the next page.

### What recognition would bring

618 ha outside counted protected areas: +0.004 points of % of Tunisia's land, of which 618 ha in KBAs not covered today. It touches a counted protected area, so it would also extend and connect the network.

Open questions: criterion A (not a protected area) is partly.

#1 · SIDI EL BARRAK DAM

# Assessment against the OECM criteria

## A · Not a protected area

PARTLY

88% is already inside counted protected areas (Barrage Sidi El Barrak (Wetland Zone of National Importance); Barrage Sidi El Barrak (Wetland of International Importance (Ramsar Site)); Dj. Chitana / Cap negro (National Park)); only the remaining 618 ha would add to 3.1.

## B · Governed and managed

LIKELY

Hunting reserve on a named site (every season): Hunting banned on the same named forest, wetland or dam in every recent hunting order (art. 12). Water Code protection zone: Water Code art. 123: polluting activities banned in the immediate (10 m) and easement (10-60 m) belts around drinking-water reservoirs. Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Legal basis: hunting order saison 2025/2026, art. 12; Code des eaux, art. 123-2; Code des eaux, art. 123-1. Authority to involve: Direction Générale des Forêts (DGF, Ministry of Agriculture) and the regional forest district (CRDA); DGBGTH (Ministry of Agriculture, dams) and SONEDE (drinking water); CRDA; DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA) (Béja governorate).

## C · Sustained in-situ biodiversity outcomes

LIKELY

100% of the net area (618 ha) lies in the KBA Maden River (Global/Regional TBD, 609 ha) and the KBA Oued El Zouara (Global/Regional TBD, 8.4 ha). Threatened trigger species of these KBAs: Anodonta lucasi (CR), Serapias stenopetala (CR), Convolvulus durandoi (CR), Calopteryx exul (EN), Pleurodeles nebulosus (VU) and Gomphus lucasii (VU). Land cover of the net area (ESA WorldCover 2021): open water 39%, tree cover 27%, grassland 22% and cropland 6%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. It touches a counted protected area (buffer and connectivity). The measure is permanent in law.

## D · Other values

MET

Game and waterbird refuge; often state forest under the Forest Code. Drinking-water security; riparian vegetation along the reservoir shore. Water supply; open water for wintering and breeding waterbirds. Supplies drinking water. Sidi el Barrak: built 2000; capacity 275 Mm3; catchment 910 km2; uses: irrigation (main); water supply.

## KBA · Maden River

Global/Regional TBD · 820.0 km² · 11% covered today. This complex covers 609 ha of it, 0.8% of its uncovered part.

Trigger species (8): Anodonta lucasi (CR); Sérapias à pétales étroits (Serapias stenopetala, CR); Liseron de durand (Convolvulus durandoi, CR); Calopteryx exul (EN); Algerian Ribbed Newt (Pleurodeles nebulosus, VU); Gomphus lucasii (VU); Bellis prostrata (NT); Tunisian Barbel (Luciobarbus callensis, LC).

Also overlaps: Oued El Zouara.

## Risks and gaps

- The order is renewed every season, so permanence rests on practice; only hunting is banned; the extent of the named area is approximate.
- Narrow belts; the law protects water quality, not biodiversity, so habitat outcomes must be shown; the mapped belt follows the reservoir outline, not the highest-water mark.
- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.
- Outline from OpenStreetMap: check it against the legal or management boundary.

## Evidence to collect

- Hunting orders over 10+ years, DGF management plan of the forest or dam, waterbird counts.
- Decree listing the protected reservoirs, enforcement, vegetation of the belt.
- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.

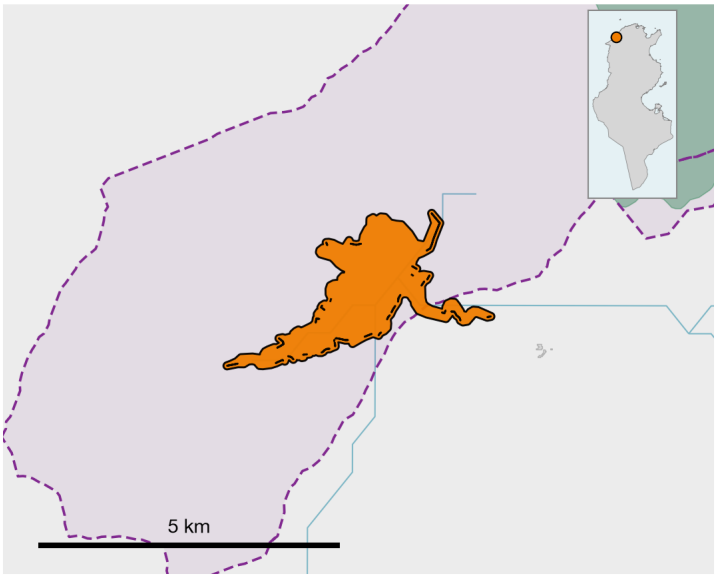
## Next steps and who

- DGF: confirm the reserve is permanent in practice and map its legal limits; consider a permanent status.
- DGBGTH / SONEDE: confirm the list of protected reservoirs and how the belt is managed.
- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- Contact: Direction Générale des Forêts; DGBGTH (Béja governorate).

PRIORITY 1 · #2 · JENDOUBA GOVERNORATE

# Ben Metir dam

Water Code protection zone · Dam reservoir or hill lake · 3 records of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**387 ha**
outside counted protected areas

**359 ha**
in KBAs (93% of the net area)

**+0.002**
points of % of Tunisia's land

**92%**
natural habitat (WorldCover)

## Why it is a priority

- In Kroumirie, KBA with 1 threatened trigger species; fills 6% of its uncovered part.
- Mostly natural habitat (open water 42%, grassland 21%, tree cover 20% and cropland 11%).
- The restriction is permanent in law.
- Its ecoregion is only 4.7% covered today.

## The site

Ben Metir dam is a dam reservoir in Jendouba governorate: built in 1954, holds 73 million m³, drains a catchment of 106 km²; it is used for irrigation first, water supply, hydropower, on the Oued El Lil. Around and on the reservoir, water code protection zone records add their own restrictions, so several measures overlap on the same ground.

None of it is counted today: 387 ha lie outside the protected areas of indicator 3.1.

## Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Kroumirie** (global or regional significance, level to be confirmed, 72.0 km², 20% covered by protected areas today).

The KBA was identified for 1 trigger species, 1 of them threatened on the IUCN Red List: Dwarf Pillwort (Pilularia minuta, EN).

Satellite land cover (ESA WorldCover 2021) shows the net area as open water 42%, grassland 21%, tree cover 20% and cropland 11%.

It belongs to the Mediterranean conifer and mixed forests ecoregion, of which only 4.7% is covered today.

## What protects it today

Water Code protection zone; Dam reservoir or hill lake.

Legal basis: Code des eaux, art. 123-1; Code des eaux, art. 123-2. Authorities: DGBGTH (Ministry of Agriculture, dams) and SONEDE (drinking water); CRDA; DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA). The full case is on the next page.

## What recognition would bring

387 ha outside counted protected areas: +0.002 points of % of Tunisia's land, of which 359 ha in KBAs not covered today.

#2 · BEN METIR DAM

# Assessment against the OECM criteria

### A · Not a protected area

MET

100% of the mapped area lies outside the protected areas and OECMs counted in indicator 3.1.

### B · Governed and managed

LIKELY

Water Code protection zone: Water Code art. 123: polluting activities banned in the immediate (10 m) and easement (10-60 m) belts around drinking-water reservoirs. Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Legal basis: Code des eaux, art. 123-2; Code des eaux, art. 123-1. Authority to involve: DGBGTH (Ministry of Agriculture, dams) and SONEDE (drinking water); CRDA; DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA) (Jendouba governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

93% of the net area (359 ha) lies in the KBA Kroumirie (Global/Regional TBD, 359 ha). That is 6% of the part of Kroumirie that no counted protected area covers today (58.1 km²; the KBA is 20% covered). Threatened trigger species of this KBA: Pilularia minuta (EN). Land cover of the net area (ESA WorldCover 2021): open water 42%, grassland 21%, tree cover 20% and cropland 11%. Ecoregion: Mediterranean conifer and mixed forests, 4.7% covered today, under-represented in the protected-area network. It lies 2.7 km from a counted protected area (connectivity). The measure is permanent in law.

### D · Other values

MET

Drinking-water security; riparian vegetation along the reservoir shore. Water supply; open water for wintering and breeding waterbirds. Supplies drinking water. Ben Metir: built 1954; dam height 78 m; capacity 73 Mm3; catchment 106 km2; uses: irrigation (main); water supply; hydropower; river El Lil.

### KBA · Kroumirie

Global/Regional TBD · 72.0 km² · 20% covered today. This complex covers 359 ha of it, 6.2% of its uncovered part.

Trigger species (1): Dwarf Pillwort (Pilularia minuta, EN).

### Risks and gaps

- Narrow belts; the law protects water quality, not biodiversity, so habitat outcomes must be shown; the mapped belt follows the reservoir outline, not the highest-water mark.
- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.
- Outline from OpenStreetMap: check it against the legal or management boundary.

### Evidence to collect

- Decree listing the protected reservoirs, enforcement, vegetation of the belt.
- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.

### Next steps and who

- DGBGTH / SONEDE: confirm the list of protected reservoirs and how the belt is managed.
- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- Contact: DGBGTH (Jendouba governorate).

### SCORECARD

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

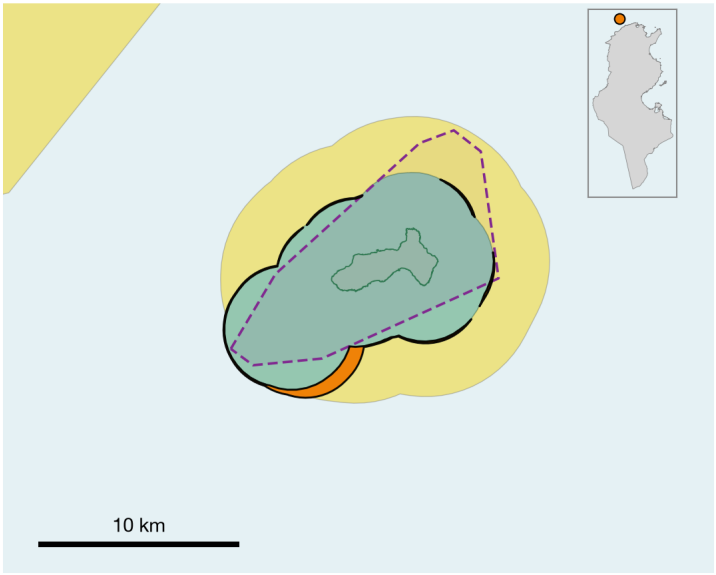



0.67
screening score

PRIORITY 1 · #3 · BIZERTE GOVERNORATE

# No-fishing zone around La Galite and Galiton (1.5 mile)

No-fishing zone · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**453 ha**
outside counted protected areas

**24 ha**
in KBAs (5% of the net area)

**+0.005**
points of % of Tunisia's sea

**0.64**
screening score (0–1)

### Why it is a priority

- In Archipel de la Galite, a KBA; fills 2% of its uncovered part.
- The restriction is permanent in law.
- Next to a counted protected area: extends and connects it.

### The site

A no-fishing zone 1.5 nautical miles wide surrounds the islands of La Galite and Galiton, off Tabarka in the north-west. It was created by the fishing order of 28 September 1995 (art. 25), which bans all fishing there all year.

Most of the mapped area (95%) already lies inside counted protected areas; only 453 ha around them would add to indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Archipel de la Galite** (regional significance, 81.4 km², 87% covered by protected areas today). Its factsheet describes it this way: “The Galite archipelago is situated 40 km off the coast of Tunisia, 45 km north-west of Cap Serrat and 64 km north-east of the town of Tabarka.”

Its trigger species include European Leaf-toed Gecko (Euleptes europaea, NT), Eleonora's Falcon (Falco eleonorae, LC), Audouin's Gull (Larus audouinii, LR/cd), Sardinian Warbler (Sylvia melanocephala, LR/lc).

The factsheet notes: “The site is likely to be declared a National Park. The seas surrounding the archipelago are a Marine Reserve”

### What protects it today

No-fishing zone. Legal basis: Arrêté du 28/09/1995, art. 25. Authorities: Direction Générale de la Pêche et de l'Aquaculture (DGPA, Ministry of Agriculture); Coast Guard for enforcement. The full case is on the next page.

### What recognition would bring

453 ha outside counted protected areas: +0.005 points of % of Tunisia's sea, of which 24 ha in KBAs not covered today. It touches a counted protected area, so it would also extend and connect the network.

Open questions: criterion A (not a protected area) is partly; criterion C (sustained in-situ biodiversity outcomes) is uncertain.

#3 · NO-FISHING ZONE AROUND LA GALITE AND GALITON (1.5 MILE)

# Assessment against the OECM criteria

### A · Not a protected area

PARTLY

95% is already inside counted protected areas (Archipel De La Galite (Specially Protected Areas of Mediterranean Importance (Barcelona Convention)); La Galite (Nature Reserve)); only the remaining 453 ha would add to 3.1.

### B · Governed and managed

LIKELY

No-fishing zone: All fishing is banned all year (fishing order of 28/09/1995, art. 25). Legal basis: Arrêté du 28/09/1995, art. 25. Authority to involve: Direction Générale de la Pêche et de l'Aquaculture (DGPA, Ministry of Agriculture); Coast Guard for enforcement (Bizerte governorate).

### C · Sustained in-situ biodiversity outcomes

UNCERTAIN

5% of the net area (24 ha) lies in the KBA Archipel de la Galite (Regional, 24 ha). That is 2% of the part of Archipel de la Galite that no counted protected area covers today (10.7 km²; the KBA is 87% covered). It touches a counted protected area (buffer and connectivity). The measure is permanent in law.

### D · Other values

MET

Refuge for fish stocks next to fishing grounds (spill-over).

### KBA · Archipel de la Galite

Regional · 81.4 km² · 87% covered today. This complex covers 24 ha of it, 2.3% of its uncovered part.

Trigger species (5): European Leaf-toed Gecko (Euleptes europaea, NT); Eleonora's Falcon (Falco eleonorae, LC); Audouin's Gull (Larus audouinii, LR/cd); Sardinian Warbler (Sylvia melanocephala, LR/lc); Bellevalia galitensis ().

### Risks and gaps

- Check that the 1995 text is still in force and enforced; most of the island zones already lie inside the national park.

### Evidence to collect

- Enforcement records, fish biomass or visual census inside vs outside.

### Next steps and who

- DGPA: confirm the text in force; report the part outside the park as an OECM, or extend the park boundary.
- Contact: Direction Générale de la Pêche et de l'Aquaculture (Bizerte governorate).

## SCORECARD

| A | B | UNCERTAIN | D |
|---|---|---|---|
| PARTLY | LIKELY | | MET |

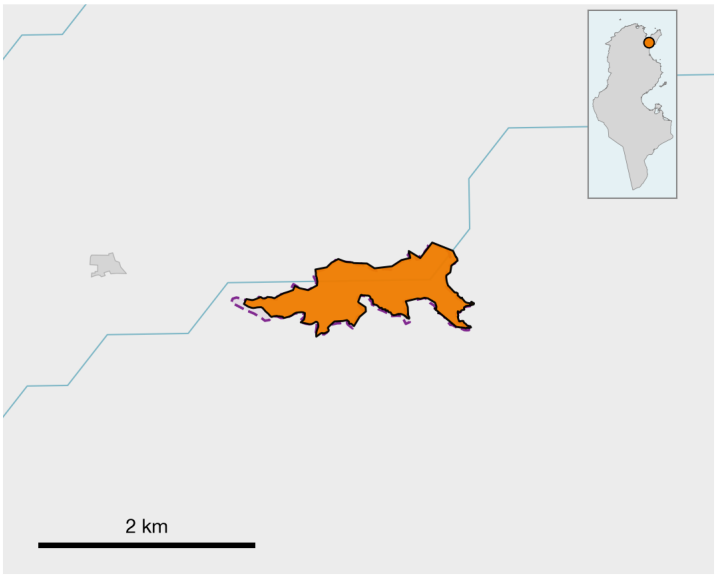



0.64
screening score

PRIORITY 1 · #4 · NABEUL GOVERNORATE

# El Masri dam

Dam reservoir or hill lake · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**77 ha**
outside counted protected areas

**72 ha**
in KBAs (94% of the net area)

**<0.001**
points of % of Tunisia's land

**99%**
natural habitat (WorldCover)

## Why it is a priority

- In Barrage Masri, KBA with 1 threatened trigger species; fills 93% of its uncovered part.
- Mostly natural habitat (open water 86% and grassland 6%).
- The restriction is permanent in law.
- Its ecoregion is only 2.4% covered today.

## The site

El Masri dam is a dam reservoir in Nabeul governorate: built in 1965, holds 7 million m³, drains a catchment of 42 km²; it is used for irrigation first.

None of it is counted today: 77 ha lie outside the protected areas of indicator 3.1.

## Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Barrage Masri** (global significance, 0.8 km², 0% covered by protected areas today). Its factsheet describes it this way: “Masri reservoir is, unlike the Cap Bon reservoirs, being constructed for water-supply to the national grid rather than local irrigation.”

The KBA was identified for 1 trigger species, 1 of them threatened on the IUCN Red List: White-headed Duck (Oxyura leucocephala, EN).

Satellite land cover (ESA WorldCover 2021) shows the net area as open water 86% and grassland 6%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

## What protects it today

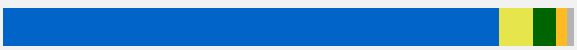
Dam reservoir or hill lake. Authorities: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA). The full case is on the next page.

## What recognition would bring

77 ha outside counted protected areas: less than 0.001 points of % of Tunisia's land, of which 72 ha in KBAs not covered today.

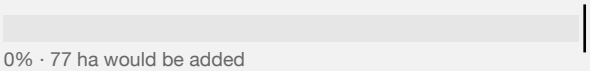
### AT A GLANCE

**Land cover (ESA WorldCover 2021)**



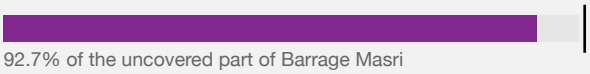
water 86% · grass 6%

**Already inside counted protected areas**



0% · 77 ha would be added

**KBA gap it would fill**



92.7% of the uncovered part of Barrage Masri

**Measures on this ground**

Dam reservoir or hill lake

#4 · EL MASRI DAM

# Assessment against the OECM criteria

### A · Not a protected area

MET

100% of the mapped area lies outside the protected areas and OECMs counted in indicator 3.1.

### B · Governed and managed

LIKELY

Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Authority to involve: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA) (Nabeul governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

94% of the net area (72 ha) lies in the KBA Barrage Masri (Global, 72 ha). That is 93% of the part of Barrage Masri that no counted protected area covers today (0.8 km²; the KBA is 0% covered). Threatened trigger species of this KBA: Oxyura leucocephala (EN). Land cover of the net area (ESA WorldCover 2021): open water 86% and grassland 6%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. The measure is permanent in law.

### D · Other values

MET

Water supply; open water for wintering and breeding waterbirds. El Masri: built 1965; dam height 38 m; capacity 7 Mm3; catchment 42 km2; uses: irrigation (main).

### KBA · Barrage Masri

Global · 0.8 km² · 0% covered today. This complex covers 72 ha of it, 92.7% of its uncovered part.

Trigger species (1): White-headed Duck (Oxyura leucocephala, EN).

### Risks and gaps

- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.
- Outline from OpenStreetMap: check it against the legal or management boundary.

### Evidence to collect

- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.

### Next steps and who

- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- Contact: DGBGTH (Nabeul governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

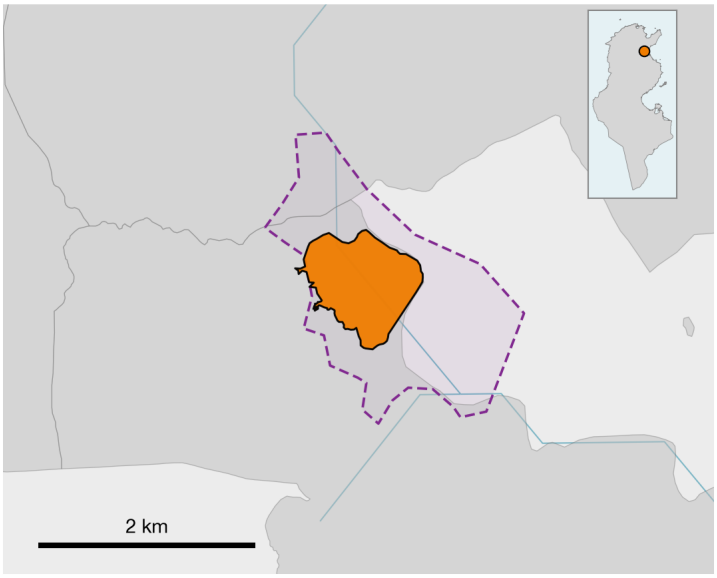



0.63
screening score

PRIORITY 1 · #5 · SOUSSE GOVERNORATE

# El Ougla dam

Dam reservoir or hill lake · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**76 ha**
outside counted protected areas

**75 ha**
in KBAs (99% of the net area)

**<0.001**
points of % of Tunisia's land

**97%**
natural habitat (WorldCover)

### Why it is a priority

- In Barrage Khairat, KBA with 2 threatened trigger species; fills 24% of its uncovered part.
- Mostly natural habitat (open water 95%).
- The restriction is permanent in law.
- Its ecoregion is only 2.4% covered today.

### The site

El Ougla dam is a dam reservoir in Sousse governorate: built in 1988, holds 5.3 million m³, drains a catchment of 164 km²; it is used for irrigation first, on the Oued El Ougla.

None of it is counted today: 76 ha lie outside the protected areas of indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Barrage Khairat** (global or regional significance, level to be confirmed, 3.2 km², 0% covered by protected areas today).

The KBA was identified for 2 trigger species, 2 of them threatened on the IUCN Red List: White-headed Duck (Oxyura leucocephala, EN), Marbled Teal (Marmaronetta angustirostris, VU).

Satellite land cover (ESA WorldCover 2021) shows the net area as open water 95%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

### What protects it today

Dam reservoir or hill lake. Authorities: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA). The full case is on the next page.

### What recognition would bring

76 ha outside counted protected areas: less than 0.001 points of % of Tunisia's land, of which 75 ha in KBAs not covered today.

#### AT A GLANCE

**Land cover (ESA WorldCover 2021)**

water 95%

**Already inside counted protected areas**

0% · 76 ha would be added

**KBA gap it would fill**

23.5% of the uncovered part of Barrage Khairat

**Measures on this ground**

Dam reservoir or hill lake

#5 · EL OUGLA DAM

# Assessment against the OECM criteria

### A · Not a protected area

MET

100% of the mapped area lies outside the protected areas and OECMs counted in indicator 3.1.

### B · Governed and managed

LIKELY

Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Authority to involve: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA) (Sousse governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

99% of the net area (75 ha) lies in the KBA Barrage Khairat (Global/Regional TBD, 75 ha). That is 24% of the part of Barrage Khairat that no counted protected area covers today (3.2 km²; the KBA is 0% covered). Threatened trigger species of this KBA: Oxyura leucocephala (EN) and Marmaronetta angustirostris (VU). Land cover of the net area (ESA WorldCover 2021): open water 95%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. The measure is permanent in law.

### D · Other values

MET

Water supply; open water for wintering and breeding water-birds. El Ougla: built 1988; dam height 19 m; capacity 5.3 Mm3; catchment 164 km2; uses: irrigation (main); river El Ougla.

### KBA · Barrage Khairat

Global/Regional TBD · 3.2 km² · 0% covered today. This complex covers 75 ha of it, 23.5% of its uncovered part.

Trigger species (2): White-headed Duck (Oxyura leucocephala, EN); Marbled Teal (Marmaronetta angustirostris, VU).

### Risks and gaps

- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.
- Outline from OpenStreetMap: check it against the legal or management boundary.

### Evidence to collect

- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.

### Next steps and who

- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- Contact: DGBGTH (Sousse governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

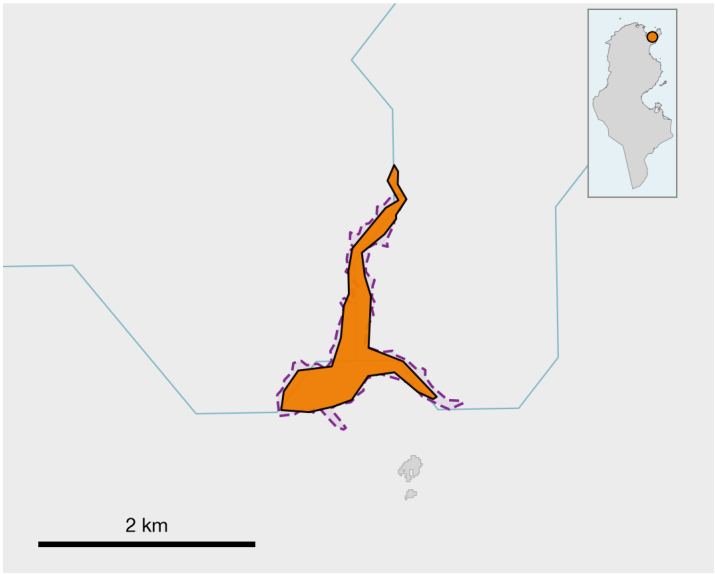



0.63
screening score

PRIORITY 1 · #6 · NABEUL GOVERNORATE

# Bezirk dam

Dam reservoir or hill lake · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**68 ha**
outside counted protected areas

**64 ha**
in KBAs (94% of the net area)

**<0.001**
points of % of Tunisia's land

**97%**
natural habitat (WorldCover)

### Why it is a priority

- In Barrage Bezikh, KBA with 1 threatened trigger species; fills 76% of its uncovered part.
- Mostly natural habitat (open water 78% and grassland 18%).
- The restriction is permanent in law.
- Its ecoregion is only 2.4% covered today.

### The site

Bezirk dam is a dam reservoir in Nabeul governorate: built in 1959, holds 6.5 million m³, drains a catchment of 69 km²; it is used for irrigation first, on the Bezirk.

None of it is counted today: 68 ha lie outside the protected areas of indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Barrage Bezikh** (global or regional significance, level to be confirmed, 0.8 km², 0% covered by protected areas today).

The KBA was identified for 1 trigger species, 1 of them threatened on the IUCN Red List: White-headed Duck (Oxyura leucocephala, EN).

Satellite land cover (ESA WorldCover 2021) shows the net area as open water 78% and grassland 18%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

### What protects it today

Dam reservoir or hill lake. Authorities: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA). The full case is on the next page.

### What recognition would bring

68 ha outside counted protected areas: less than 0.001 points of % of Tunisia's land, of which 64 ha in KBAs not covered today.

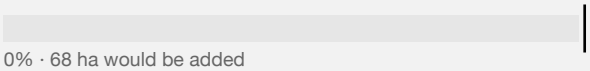
#### AT A GLANCE

**Land cover (ESA WorldCover 2021)**



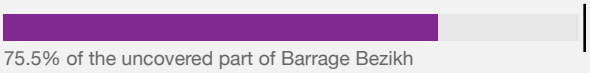
water 78% · grass 18%

**Already inside counted protected areas**



0% · 68 ha would be added

**KBA gap it would fill**



75.5% of the uncovered part of Barrage Bezikh

**Measures on this ground**

Dam reservoir or hill lake

#6 · BEZIRK DAM

# Assessment against the OECM criteria

### A · Not a protected area

MET

100% of the mapped area lies outside the protected areas and OECMs counted in indicator 3.1.

### B · Governed and managed

LIKELY

Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Authority to involve: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA) (Nabeul governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

94% of the net area (64 ha) lies in the KBA Barrage Bezikh (Global/Regional TBD, 64 ha). That is 76% of the part of Barrage Bezikh that no counted protected area covers today (0.8 km²; the KBA is 0% covered). Threatened trigger species of this KBA: Oxyura leucocephala (EN). Land cover of the net area (ESA WorldCover 2021): open water 78% and grassland 18%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. The measure is permanent in law.

### D · Other values

MET

Water supply; open water for wintering and breeding waterbirds. Bezirk: built 1959; dam height 24 m; capacity 6.5 Mm3; catchment 69 km2; uses: irrigation (main); river Bezirk.

### KBA · Barrage Bezikh

Global/Regional TBD · 0.8 km² · 0% covered today. This complex covers 64 ha of it, 75.5% of its uncovered part.

Trigger species (1): White-headed Duck (Oxyura leucocephala, EN).

### Risks and gaps

- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.

### Evidence to collect

- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.

### Next steps and who

- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- Contact: DGBGTH (Nabeul governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

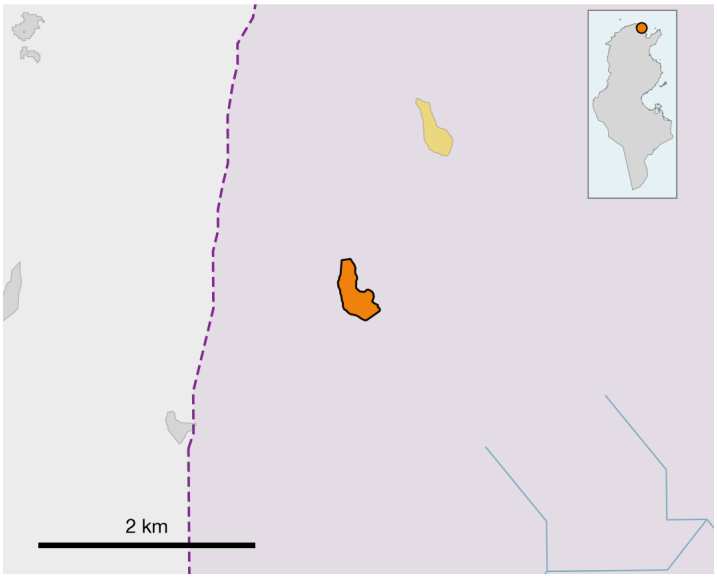



**0.63**
screening score

PRIORITY 1 · #7 · BIZERTE GOVERNORATE

# Dam reservoir or hill lake, Bizerte (GDW 28423)

Dam reservoir or hill lake · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**11 ha**
outside counted protected areas

**11 ha**
in KBAs (100% of the net area)

**<0.001**
points of % of Tunisia's land

**98%**
natural habitat (WorldCover)

### Why it is a priority

- In Jbel Nadhour et Lagune de Ghar El Melh, KBA with 1 threatened trigger species.
- Mostly natural habitat (open water 87% and grassland 8%).
- The restriction is permanent in law.
- Its ecoregion is only 2.4% covered today.

### The site

Dam reservoir or hill lake, Bizerte (GDW 28423) is a dam reservoir in Bizerte governorate: built in 1985, holds 1.4 million m³, drains a catchment of 8 km².

None of it is counted today: 11 ha lie outside the protected areas of indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Jbel Nadhour et Lagune de Ghar El Melh** (global or regional significance, level to be confirmed, 239.5 km², 16% covered by protected areas today).

The KBA was identified for 1 trigger species, 1 of them threatened on the IUCN Red List: Blanc's Fringe-toed Lizard (Acanthodactylus blanci, EN).

Satellite land cover (ESA WorldCover 2021) shows the net area as open water 87% and grassland 8%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

### What protects it today

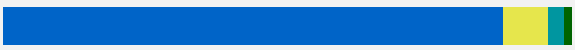
Dam reservoir or hill lake. Authorities: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA). The full case is on the next page.

### What recognition would bring

11 ha outside counted protected areas: less than 0.001 points of % of Tunisia's land, of which 11 ha in KBAs not covered today.

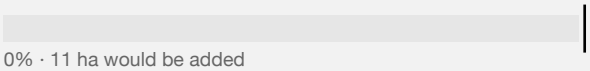
#### AT A GLANCE

**Land cover (ESA WorldCover 2021)**



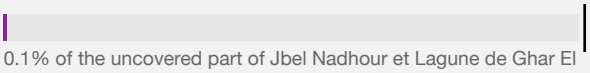
water 87% · grass 8%

**Already inside counted protected areas**



0% · 11 ha would be added

**KBA gap it would fill**



0.1% of the uncovered part of Jbel Nadhour et Lagune de Ghar El Melh

**Measures on this ground**

Dam reservoir or hill lake

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

#7 · DAM RESERVOIR OR HILL LAKE, BIZERTE (GDW 28423)

# Assessment against the OECM criteria

### A · Not a protected area

MET

100% of the mapped area lies outside the protected areas and OECMs counted in indicator 3.1.

### B · Governed and managed

LIKELY

Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Authority to involve: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA) (Bizerte governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

100% of the net area (11 ha) lies in the KBA Jbel Nadhour et Lagune de Ghar El Melh (Global/Regional TBD, 11 ha). Threatened trigger species of this KBA: Acanthodactylus blanci (EN). Land cover of the net area (ESA WorldCover 2021): open water 87% and grassland 8%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. It lies 3.8 km from a counted protected area (connectivity). The measure is permanent in law.

### D · Other values

MET

Water supply; open water for wintering and breeding waterbirds. Dam reservoir or hill lake, Bizerte (GDW 28423): built 1985; capacity 1.4 Mm3; catchment 8 km2.

### KBA · Jbel Nadhour et Lagune de Ghar El Melh

Global/Regional TBD · 239.5 km² · 16% covered today. This complex covers 11 ha of it, 0.1% of its uncovered part.

Trigger species (1): Blanc's Fringe-toed Lizard (Acanthodactylus blanci, EN).

### Risks and gaps

- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.
- Outline from OpenStreetMap: check it against the legal or management boundary.

### Evidence to collect

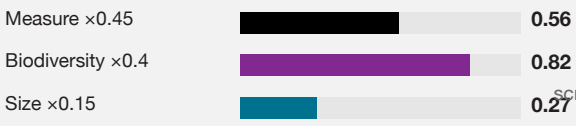
- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.

### Next steps and who

- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- Contact: DGBGTH (Bizerte governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

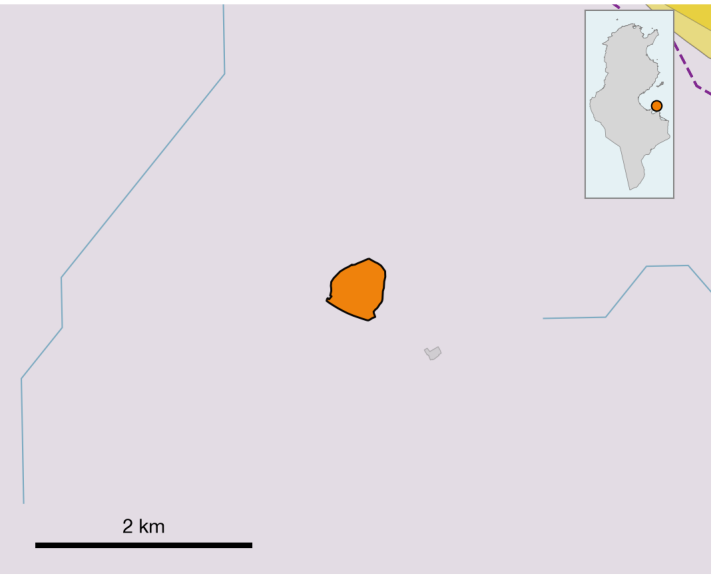



**0.62**
screening score

PRIORITY 1 · #8 · MÉDENINE GOVERNORATE

# Henchir Bourgou

Archaeological site · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**21 ha**
outside counted protected areas

**21 ha**
in KBAs (100% of the net area)

**<0.001**
points of % of Tunisia's land

**82%**
natural habitat (WorldCover)

### Why it is a priority

- In Île de Djerba, a KBA.
- Mostly natural habitat (grassland 74% and bare / sparse vegetation 25%).
- The restriction is permanent in law.
- Its ecoregion is only 2.4% covered today.

### The site

Henchir Bourgou is an archaeological site in Médenine governorate, on the island of Djerba. Under the Heritage Code (Law 94-35), construction, digging and changes of use on archaeological sites need the approval of the Institut National du Patrimoine, which, with the AM-VPPC, manages the sites.

None of it is counted today: 21 ha lie outside the protected areas of indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Île de Djerba** (global or regional significance, level to be confirmed, 484.1 km², 2% covered by protected areas today).

Satellite land cover (ESA WorldCover 2021) shows the net area as grassland 74% and bare / sparse vegetation 25%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

### What protects it today

Archaeological site. Authorities: Institut National du Patrimoine (INP) and AMVPPC (site management). The full case is on the next page.

### What recognition would bring

21 ha outside counted protected areas: less than 0.001 points of % of Tunisia's land, of which 21 ha in KBAs not covered today.

#### AT A GLANCE

**Land cover (ESA WorldCover 2021)**

grass 74% · bare 25%

**Already inside counted protected areas**

0% · 21 ha would be added

**KBA gap it would fill**

0.0% of the uncovered part of Île de Djerba

**Measures on this ground**

Archaeological site

#8 · HENCHIR BOURGOU

# Assessment against the OECM criteria

### A · Not a protected area

MET

100% of the mapped area lies outside the protected areas and OECMs counted in indicator 3.1.

### B · Governed and managed

LIKELY

Archaeological site: Heritage Code (Law 94-35): construction, digging and changes need INP approval; listed sites have a protection perimeter. Authority to involve: Institut National du Patrimoine (INP) and AMVPPC (site management) (Médenine governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

100% of the net area (21 ha) lies in the KBA Île de Djerba (Global/Regional TBD, 21 ha). Land cover of the net area (ESA WorldCover 2021): grassland 74% and bare / sparse vegetation 25%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. The measure is permanent in law.

### D · Other values

MET

Archaeological and landscape heritage; tourism.

### KBA · Île de Djerba

Global/Regional TBD · 484.1 km² · 2% covered today. This complex covers 21 ha of it, 0.0% of its uncovered part.

No trigger species listed.

### Risks and gaps

- Managed for heritage, not biodiversity: vegetation clearance for visitors is common; mapped outlines (OSM) may not match the legal perimeter.
- Outline from OpenStreetMap: check it against the legal or management boundary.

### Evidence to collect

- Legal perimeter, vegetation condition (garrigue, old olive groves), INP management plan.

### Next steps and who

- INP / AMVPPC: official perimeter and vegetation management; pair biodiversity surveys with heritage monitoring.
- Contact: Institut National du Patrimoine (Médenine governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| MET | LIKELY | LIKELY | MET |

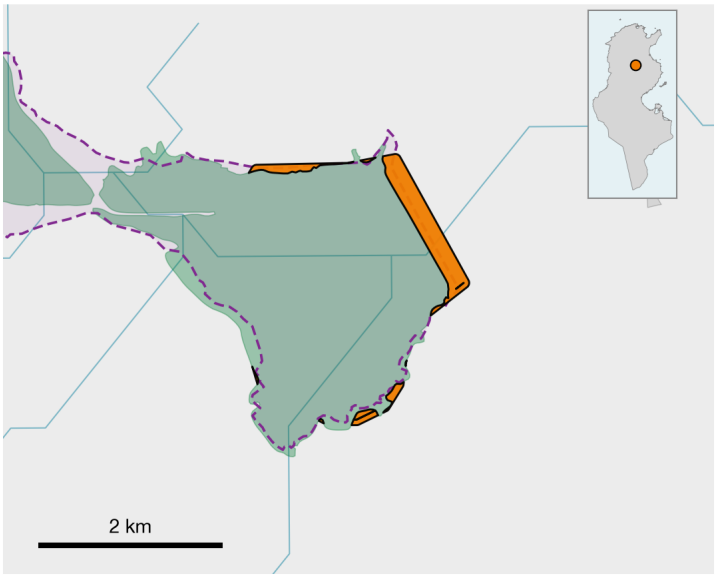



**0.61**
screening score

PRIORITY 1 · #9 · KAIROUAN GOVERNORATE

# El Habib dam

Dam reservoir or hill lake · Water Code protection zone · 3 records of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**50 ha**
outside counted protected areas

**19 ha**
in KBAs (37% of the net area)

**<0.001**
points of % of Tunisia's land

**78%**
natural habitat (WorldCover)

### Why it is a priority

- In Barrage El Houareb, KBA with 3 threatened trigger species; fills 18% of its uncovered part.
- Mostly natural habitat (bare / sparse vegetation 44%, grassland 31% and tree cover 18%).
- The restriction is permanent in law.
- Next to a counted protected area: extends and connects it.

### The site

El Habib dam is a dam reservoir in Kairouan governorate: built in 1988, holds 110 million m³, drains a catchment of 1,177 km²; it is used for irrigation first, water supply, flood control, on the Merguelil. Around and on the reservoir, water code protection zone records add their own restrictions, so several measures overlap on the same ground.

Most of the mapped area (91%) already lies inside counted protected areas; only 50 ha around them would add to indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Barrage El Houareb** (global significance, 8.7 km², 88% covered by protected areas today). Its factsheet describes it this way: “El Houareb reservoir is a man-made water-body built about 20 years ago on the Oued Merguellil, 35 km east of Kairouan, for flood-control and water-supply purposes.”

The KBA was identified for 3 trigger species, 3 of them threatened on the IUCN Red List: White-headed Duck (Oxyura leucocephala, EN), Ferruginous Duck (Aythya nyroca, VU), Marbled Teal (Marmaronetta angustirostris, VU).

The factsheet notes: “Uncontrolled hunting is the main threat to waterbirds. There are no obvious threats to the habitat and the site is patrolled by waterworks staff.”

Satellite land cover (ESA WorldCover 2021) shows the net area as bare / sparse vegetation 44%, grassland 31% and tree cover 18%.

It belongs to the Mediterranean woodlands and forests ecoregion, of which only 2.4% is covered today.

### What protects it today

Dam reservoir or hill lake; Water Code protection zone. Legal basis: Code des eaux, art. 123-1; Code des eaux, art. 123-2. Authorities: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA); DGBGTH (Ministry of Agriculture, dams) and SONEDE (drinking water); CRDA. The full case is on the next page.

### What recognition would bring

50 ha outside counted protected areas: less than 0.001 points of % of Tunisia's land, of which 19 ha in KBAs not covered today. It touches a counted protected area, so it would also extend and connect the network.

Open questions: criterion A (not a protected area) is partly.

#9 · EL HABIB DAM

# Assessment against the OECM criteria

## A · Not a protected area

PARTLY

91% is already inside counted protected areas (Barrage Merguellil - El Houareb (Wetland Zone of National Importance); Barrage Merguellil - El Houareb (Wetland of International Importance (Ramsar Site))); only the remaining 50 ha would add to 3.1.

## B · Governed and managed

LIKELY

Dam reservoir or hill lake: State-owned reservoir managed for water supply and irrigation; fishing and access are regulated. Water Code protection zone: Water Code art. 123: polluting activities banned in the immediate (10 m) and easement (10-60 m) belts around drinking-water reservoirs. Legal basis: Code des eaux, art. 123-1; Code des eaux, art. 123-2. Authority to involve: DGBGTH (Ministry of Agriculture) and the regional agriculture commissariat (CRDA); DGBGTH (Ministry of Agriculture, dams) and SONEDE (drinking water); CRDA (Kairouan governorate).

## C · Sustained in-situ biodiversity outcomes

LIKELY

37% of the net area (19 ha) lies in the KBA Barrage El Houareb (Global, 19 ha). That is 18% of the part of Barrage El Houareb that no counted protected area covers today (1.0 km²; the KBA is 88% covered). Threatened trigger species of this KBA: Oxyura leucocephala (EN), Aythya nyroca (VU) and Marmaronetta angustirostris (VU). Land cover of the net area (ESA WorldCover 2021): bare / sparse vegetation 44%, grassland 31% and tree cover 18%. Ecoregion: Mediterranean woodlands and forests, 2.4% covered today, under-represented in the protected-area network. It touches a counted protected area (buffer and connectivity). The measure is permanent in law.

## D · Other values

MET

Water supply; open water for wintering and breeding waterbirds. Drinking-water security; riparian vegetation along the reservoir shore. Supplies drinking water. El Habib: built 1988; dam height 33 m; capacity 110 Mm3; catchment 1,177 km2; uses: irrigation (main); water supply; flood control; river Merguelil.

## KBA · Barrage El Houareb

Global · 8.7 km² · 88% covered today. This complex covers 19 ha of it, 18.4% of its uncovered part.

Trigger species (3): White-headed Duck (Oxyura leucocephala, EN); Ferruginous Duck (Aythya nyroca, VU); Marbled Teal (Marmaronetta angustirostris, VU).

## Risks and gaps

- Managed for water, not biodiversity: water-level changes and fishing; outcomes must be shown with waterbird data.
- Narrow belts; the law protects water quality, not biodiversity, so habitat outcomes must be shown; the mapped belt follows the reservoir outline, not the highest-water mark.

## Evidence to collect

- International Waterbird Census counts (AAO/BirdLife Tunisia, DGF), water-level regime, fishing concessions.
- Decree listing the protected reservoirs, enforcement, vegetation of the belt.

## Next steps and who

- DGBGTH and DGF: compile waterbird counts and agree management of the shores (grazing, disturbance).
- DGBGTH / SONEDE: confirm the list of protected reservoirs and how the belt is managed.
- Contact: DGBGTH (Kairouan governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| PARTLY | LIKELY | LIKELY | MET |

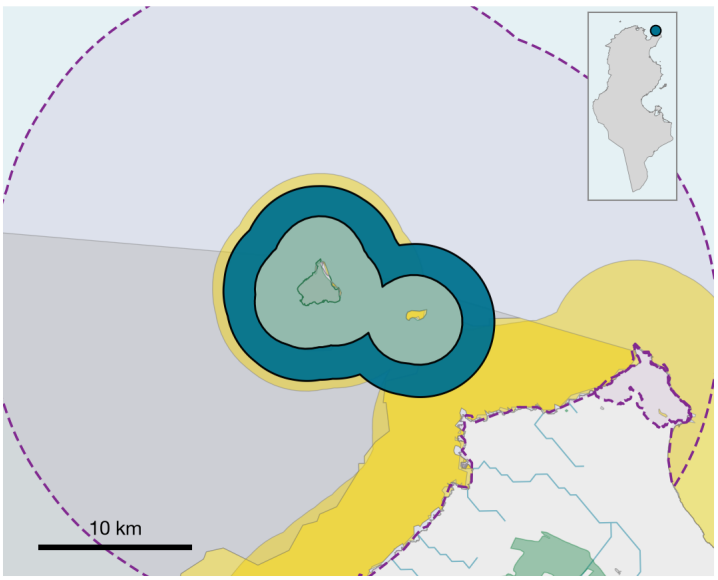



**0.60**
screening score

REPORTING GAP · #10 · NABEUL GOVERNORATE

# Parc National de Zembra et Zembretta - zone C, Transition and sustainable development area

Marine protected area zone not counted · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**91.3 km²**
outside counted protected areas

**91.3 km²**
in KBAs (100% of the net area)

**+0.091**
points of % of Tunisia's sea

**0.76**
screening score (0–1)

### Why it is a priority

- In Archipel de Zembra, a KBA; fills 7% of its uncovered part.
- The restriction is permanent in law.
- Next to a counted protected area: extends and connects it.
- Large: 91.3 km², +0.091 points of the sea indicator.

### The site

Parc National de Zembra et Zembretta - zone C, Transition and sustainable development area is a marine zone of a protected area in Nabeul governorate, recorded in the ProtectedSeas Navigator but not among the WDPA records counted in indicator 3.1. Its stated purpose is “To promote the exploitation of natural resources, and in which local populations, management bodies, scientific bodies, non-governmental organizations, cultural groups, economic interest groups and other partners cooperate to sustainably ma”. Fishing rules there: level of fishing protection 3/5; bottom trawling restricted.

Most of the mapped area (10%) already lies inside counted protected areas; only 91.3 km² around them would add to indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Archipel de Zembra** (global significance, 1,412.2 km², 6% covered by protected areas today). Its factsheet describes it this way: “Located in the eastern part of the Gulf of Tunis, to the west of Cap Bon, Zembra island is situated about 15 km west-north-west of the town of El Haouaria.”

Its trigger species include Audouin's Gull (Larus audouinii, NT), Cory's Shearwater (Calonectris diomedea, LC), Scopoli's Shearwater (Calonectris diomedea, LC).

The factsheet notes: “Zembra was declared a National Park in 1977. The site is also a Biosphere Reserve”

### What protects it today

Marine protected area zone not counted. Legal basis: National Park. Authorities: APAL (Agence de Protection et d'Aménagement du Littoral) / Ministry of Environment; DGF for national parks. The full case is on the next page.

### What recognition would bring

91.3 km² outside counted protected areas: +0.091 points of % of Tunisia's sea, of which 91.3 km² in KBAs not covered today. It touches a counted protected area, so it would also extend and connect the network.

Open questions: criterion A (not a protected area) is partly.

#10 · PARC NATIONAL DE ZEMBRA ET ZEMBRETTA - ZONE C, TRANSITION AN

# Assessment against the OECM criteria

### A · Not a protected area

PARTLY

10% is already inside counted protected areas (Zembra et Zembretta (Specially Protected Areas of Mediterranean Importance (Barcelona Convention))); only the remaining 91.3 km² would add to 3.1. ProtectedSeas records it as a marine protected area zone: report it to the WDPA as a protected area rather than as an OECM.

### B · Governed and managed

LIKELY

Marine protected area zone not counted: Marine park or reserve zone recorded by ProtectedSeas, missing from the WDPA records counted in 3.1. Legal basis: National Park. Authority to involve: APAL (Agence de Protection et d'Aménagement du Littoral) / Ministry of Environment; DGF for national parks (Nabeul governorate).

### C · Sustained in-situ biodiversity outcomes

LIKELY

100% of the net area (91.3 km²) lies in the KBA Archipel de Zembra (Global, 91.3 km²). That is 7% of the part of Archipel de Zembra that no counted protected area covers today (1,333.9 km²; the KBA is 6% covered). It touches a counted protected area (buffer and connectivity). The measure is permanent in law.

### D · Other values

MET

Island and coastal marine habitats managed for conservation.

### KBA · Archipel de Zembra

Global · 1,412.2 km² · 6% covered today. This complex covers 91.3 km² of it, 6.8% of its uncovered part.

Trigger species (3): Audouin's Gull (Larus audouinii, NT); Cory's Shearwater (Calonectris diomedea, LC); Scopoli's Shearwater (Calonectris diomedea, LC).

### Risks and gaps

- Probably a protected area rather than an OECM: the fix is to report it to the WDPA.

### Evidence to collect

- Legal text and boundary of the marine zone.

### Next steps and who

- Ministry of Environment / APAL: submit the zone to the WDPA through UNEP-WCMC (quickest gain, no new measure).
- Contact: APAL (Nabeul governorate).

## SCORECARD

| A | B | C | D |
|---|---|---|---|
| PARTLY | LIKELY | LIKELY | MET |

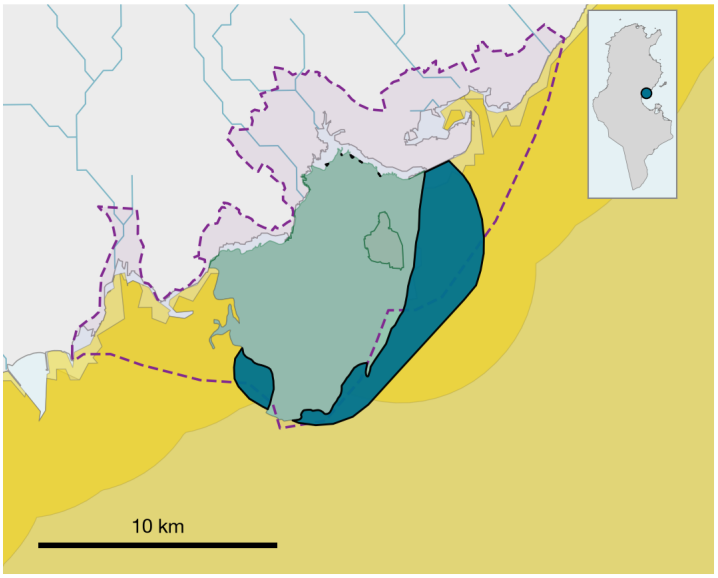



0.76
screening score

REPORTING GAP · #11 · SFAX GOVERNORATE

# Iles Kneiss

Marine protected area zone not counted · 1 record of the same place




Black outline: the complex (net of counted protected areas). Green: counted protected areas. Dashed berry: KBAs. Yellow/grey: other candidates. Blue lines: rivers.

**25.9 km²**
outside counted protected areas

**16.0 km²**
in KBAs (62% of the net area)

**+0.026**
points of % of Tunisia's sea

**0.74**
screening score (0–1)

### Why it is a priority

- In Îles Kneiss, a KBA; fills 16% of its uncovered part.
- The restriction is permanent in law.
- Next to a counted protected area: extends and connects it.

### The site

Iles Kneiss is a marine zone of a protected area in Sfax governorate, recorded in the ProtectedSeas Navigator but not among the WDPA records counted in indicator 3.1. Its stated purpose is “To preserve and rehabilitate a unique heritage of Mediterranean importance, to ensure its sustainability, and to promote its benefits for those who live around or come to visit it”. Fishing rules there: level of fishing protection 2/5; bottom trawling allowed.

Most of the mapped area (63%) already lies inside counted protected areas; only 25.9 km² around them would add to indicator 3.1.

### Why it matters for biodiversity

The complex lies in the Key Biodiversity Area **Îles Kneiss** (global significance, 159.3 km², 36% covered by protected areas today). Its factsheet describes it this way: “Located 50 km south of the town of Sfax and 25 km south-east of Mahares, Kneiss is a large area of wetland in the Gulf of Gabès.”

Its trigger species include Eurasian Oystercatcher (Haematopus ostralegus, NT), Black-tailed Godwit (Limosa limosa, NT), Dunlin (Calidris alpina, LC), Red-necked Nightjar (Caprimulgus ruficollis, LC).

The factsheet notes: “The site was declared a Natural Reserve by ministerial order in 1993”

### What protects it today

Marine protected area zone not counted. Legal basis: Nature Reserve; Specially Protected Area of Mediterranean Importance (Barcelona Convention); Ramsar site, Wetland of International Importance; Wetland Zone of National Importance. Authorities: APAL (Agence de Protection et d'Aménagement du Littoral) / Ministry of Environment; DGF for national parks. The full case is on the next page.

### What recognition would bring

25.9 km² outside counted protected areas: +0.026 points of % of Tunisia's sea, of which 16.0 km² in KBAs not covered today. It touches a counted protected area, so it would also extend and connect the network.

Open questions: criterion A (not a protected area) is partly.

# Assessment against the OECM criteria

## A · Not a protected area

PARTLY

63% is already inside counted protected areas (Kneiss iles (Specially Protected Areas of Mediterranean Importance (Barcelona Convention)); Kneiss iles (Wetland Zone of National Importance); Kneiss iles (Wetland of International Importance (Ramsar Site))); only the remaining 25.9 km² would add to 3.1. ProtectedSeas records it as a marine protected area zone: report it to the WDPA as a protected area rather than as an OECM.

## B · Governed and managed

LIKELY

Marine protected area zone not counted: Marine park or reserve zone recorded by ProtectedSeas, missing from the WDPA records counted in 3.1. Legal basis: Nature Reserve; Specially Protected Area of Mediterranean Importance (Barcelona Convention); Ramsar site, Wetland of International Importance; Wetland Zone of National Importance. Authority to involve: APAL (Agence de Protection et d'Aménagement du Littoral) / Ministry of Environment; DGF for national parks (Sfax governorate).

## C · Sustained in-situ biodiversity outcomes

LIKELY

62% of the net area (16.0 km²) lies in the KBA Îles Kneiss (Global, 16.0 km²). That is 16% of the part of Îles Kneiss that no counted protected area covers today (101.5 km²; the KBA is 36% covered). It touches a counted protected area (buffer and connectivity). The measure is permanent in law.

## D · Other values

MET

Island and coastal marine habitats managed for conservation.

## KBA · Îles Kneiss

Global · 159.3 km² · 36% covered today. This complex covers 16.0 km² of it, 15.8% of its uncovered part.

Trigger species (29): Eurasian Oystercatcher (Haematopus ostralegus, NT); Black-tailed Godwit (Limosa limosa, NT); Dunlin (Calidris alpina, LC); Red-necked Nightjar (Caprimulgus ruficollis, LC); Kentish Plover (Charadrius alexandrinus, LC); Common Ringed Plover (Charadrius hiaticula, LC); Temminck's Lark (Eremophila bilopha, LC); Black-eared Wheatear (Oenanthe hispanica, LC); Black Wheatear (Oenanthe leucura, LC); Buff-rumped Wheatear (Oenanthe moesta, LC); and 19 more.

## Risks and gaps

- Probably a protected area rather than an OECM: the fix is to report it to the WDPA.

## Evidence to collect

- Legal text and boundary of the marine zone.

## Next steps and who

- Ministry of Environment / APAL: submit the zone to the WDPA through UNEP-WCMC (quickest gain, no new measure).
- Contact: APAL (Sfax governorate).

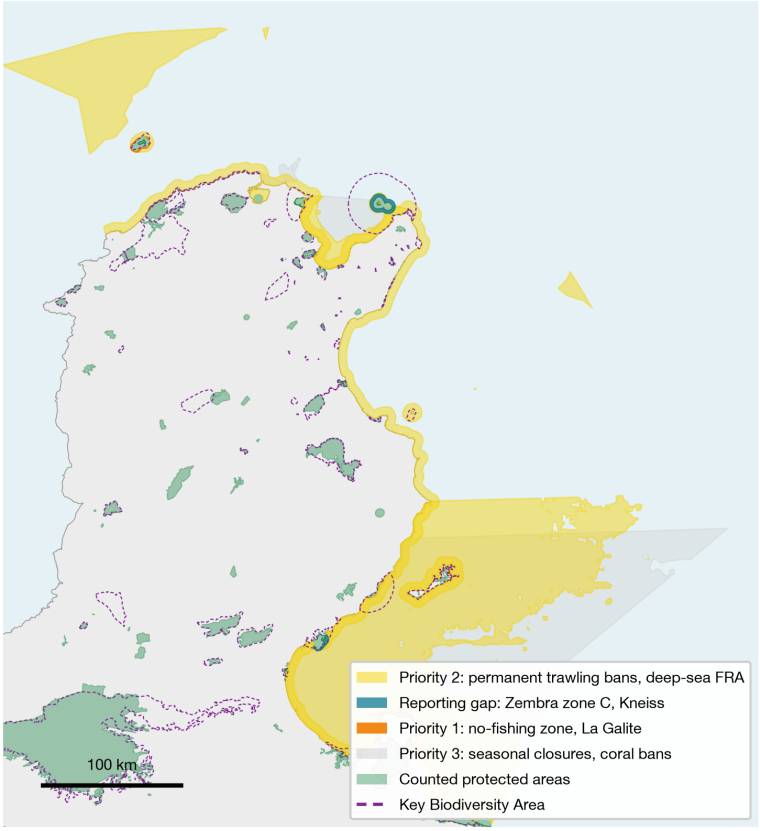
## SCORECARD

| A | B | C | D |
|---|---|---|---|
| PARTLY | LIKELY | LIKELY | MET |




0.74
screening score

# The trawling bans decide 30% at sea




Fisheries candidates by priority, net of counted protected areas. Dashed: KBAs.

Sea coverage counted today is **1.07%**, 28,926 km² short of 30%. Three kinds of marine measure stand out.

### Report the protected zones

Zembra–Zembretta zone C and the Kneiss islands (117 km²) are marine park and reserve zones missing from the WDPA. They are protected areas: reporting them is an administrative step.

### Assess the permanent trawling bans

Bottom trawling is banned all year in the Gulf of Gabès and the Gulf of Tunis under 50 m and within 3 miles of the coast (fishing order of 28/09/1995), and below 1,000 m across the GFCM deep-sea restricted area. Together they would take sea coverage to about **34%**.

They are priority 2, not 1: only one gear is banned, and illegal trawling in shallow waters is widely reported in the Gulf of Gabès. FAO and CBD guidance accept fisheries measures as OECMs when they deliver biodiversity outcomes: Tunisia would need to show seagrass condition and enforcement.

### Leave the seasonal closures as context

The July–September trawling closure of the Gulf of Gabès and the coral-harvesting bans do not give year-round, long-term protection.

| Priority-2 marine measure | Net km² | + points (sea) | In KBAs km² |
|---|---|---|---|
| Gulf of Gabès trawling ban (depths < 50 m) | 23,514 | 23.52 | 468 |
| Coastal 3-mile trawling ban | 6,947 | 6.95 | 856 |
| GFCM FRA - Deepwater FRA (below 1000 m) | 5,031 | 5.03 | 0 |
| Gulf of Tunis trawling ban (depths < 50 m) | 1,018 | 1.02 | 166 |
| No-fishing zones around offshore oil installations (500 m) | 1 | 0.00 | 0 |

## WHAT IS AT STAKE WITH THE TRAWLING BANS

**36,510 km²**
of sea under permanent trawling bans

**+32.9**
points at sea if they qualify

**1,490 km²**
of KBA they overlap

**1 gear**
banned: bottom trawling only

**Evidence that would make the case**

VMS / AIS tracks of trawlers · patrol and offence records · seagrass extent and condition · fish and nursery surveys (INSTM) · small-scale fishers' observations · GFCM reporting

# Priority 2: 97 complexes to assess case by case

## Promising, but each needs a decision, evidence or a narrower boundary first.

The 97 priority-2 complexes score above the threshold but fall short of priority 1 for a specific reason: a low-confidence outline (sacred groves), a measure that bans a single activity (trawling bans, whole-imada hunting reserves), no KBA (most reservoirs), or a farmed system whose biodiversity must be documented (oases). They are grouped below by what holds them back; each group needs a different next step.

### Sacred groves (low-confidence outlines) · 13

Wooded shrines, mostly in KBAs of the Mogods and Kroumirie. Check each patch on imagery and with the custodians; recognition needs community consent.

Grove at Sidi Abd el Kader (Béja, 10 ha); Grove at Sidi Gnaoui (Béja, 7 ha); Grove at Sidi Mannsour (Béja, 6 ha); Grove at Sidi Mahmed el Djemil (Béja, 11 ha); Grove at Sidi Daou (Béja, 5 ha); Grove at Sidi Marklouf (Béja, 20 ha); Grove at Sidi Bou Ihren (Bizerte, 11 ha); Grove at Sidi Salem (Bizerte, 10 ha); Grove at Sidi Abd el Hafid (Zaghouan, 10 ha); Grove at Sidi Jmra (El Kef, 6 ha); Grove at Sidi Abd el Kader (Kasserine, 9 ha); Grove at Sidi Ahmed Baleus (Bizerte, 9 ha); Grove at Sidi Ladhour (Bizerte, 7 ha).

### Dams and hill lakes · 25

Reservoirs in or near KBAs, or large reservoirs outside KBAs (Sidi Salem, Nebeur, Sejnane). Waterbird counts decide.

Dam reservoir or hill lake, Nabeul (GDW 16550) (Nabeul, 37 ha); Dam reservoir or hill lake, Zaghouan (GDW 16561) (Zaghouan, 61 ha); Dam reservoir or hill lake, Béja (Béja, 13 ha); Lebna dam (Nabeul, 256 ha); Dam reservoir or hill lake, Bizerte (GDW 32607) (Bizerte, 8 ha); Dam reservoir or hill lake, Sousse (Sousse, 12 ha); Dam reservoir or hill lake, Nabeul (GDW 31849) (Nabeul, 9 ha); Chiba dam (Nabeul, 125 ha); Dam reservoir or hill lake, Béja (Béja, 6 ha); Dam reservoir or hill lake, Nabeul (Nabeul, 12 ha); Mornaguia dam (Tunis, 31 ha); Sidi Salem dam (Béja, 48.5 km²); Nebeur dam (El Kef, 515 ha); Siliana dam (Siliana, 13.8 km²) and 11 more.

### Traditional oases · 23

Safeguard zones under Law 83-87, many inside the Chott Djerid KBA outline. OECM status rests on agro-biodiversity and traditional water governance.

Oasis of Beni M'hamed (Kébili, 662 ha); Oasis of Djemna (Kébili, 24.5 km²); Oasis of Barghouthia (Kébili, 611 ha); Oases of El Golâa - Douz (Kébili, 54.3 km²); Oasis of Guettaya (Kébili, 269 ha); Oasis of Blidet (Kébili, 215 ha); Imadat Âlaya (Gabès, 48.2 km²); Oasis of Sedghiene (Médenine, 28 ha); Traditional oasis, Gafsa (Gafsa, 908 ha); Oasis of Kebili (Kébili, 10.4 km²); Oasis of Negga (Kébili, 402 ha); Oasis of Zarcine (Kébili, 10.5 km²); Traditional oasis, Kébili (Kébili, 15.0 km²); Gafsa oasis (Gafsa, 17.6 km²) and 9 more.

### Hunting reserves · 15

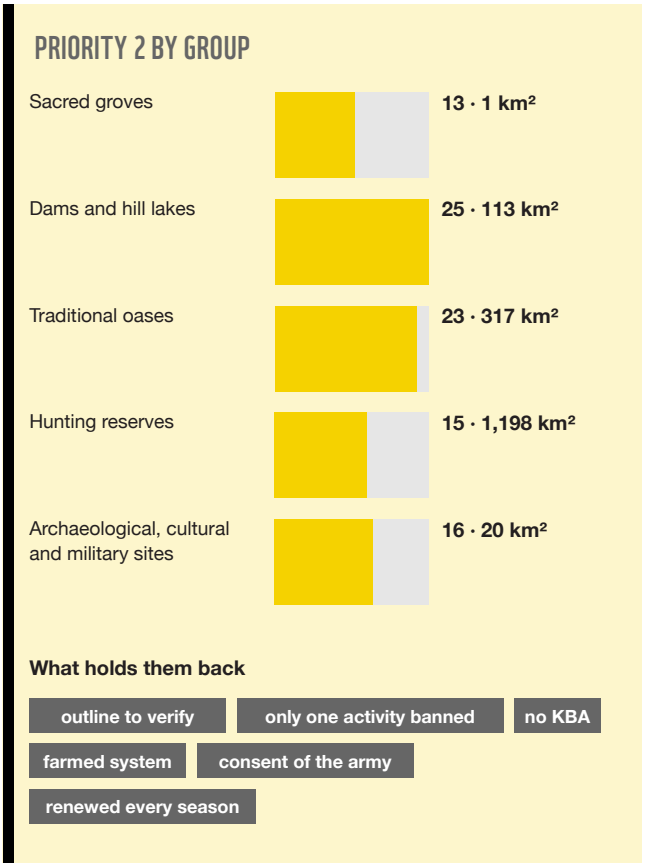
Reserves listed every season; whole imadas are capped at priority 2 because they include villages and farmland.

Imadat Bou Abdella (Siliana, 47.3 km²); Imadat Ettalah Est y compris la Réserve Naturelle de Haddaj (Gafsa, 113.8 km²); Imadat Bchimet Elborej (Gabès, 49.9 km²); Imadat Hbabssa Sud (Siliana, 99.2 km²); Imadat Mansoura Nord (Siliana, 36.9 km²); Imadat Touila (Kairouan, 105.7 km²); Imadat El Mhamela (Gabès, 264.1 km²); Imadat Tebedit (Gafsa, 84.7 km²); Forêt El Âacheba (Mahdia, 247 ha); Imadat El Médou (Gabès, 114.3 km²); Imadat Aîn Boussâdia (Siliana, 56.4 km²); Imadat Sidi Hamada (Siliana, 36.4 km²); Imadat Khmouda Nord (Kasserine, 53.2 km²); Imadat El Âaribet (Sousse, 23.7 km²) and 1 more.

### Archaeological, cultural and military sites · 16

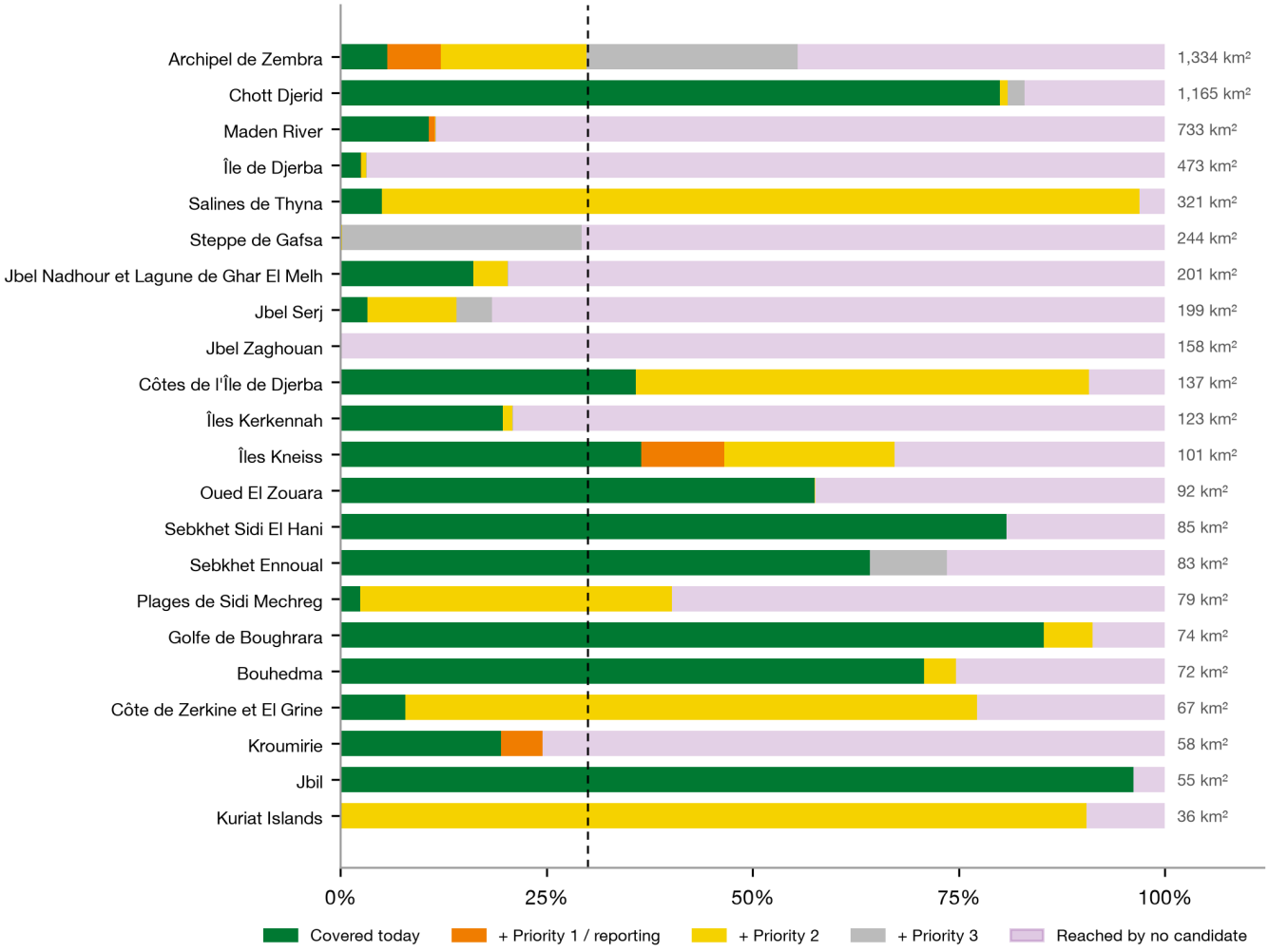
Permanent and managed; their vegetation is the question. Military sites need the Ministry of Defence's consent.

Military area, Gafsa (Gafsa, 671 ha); Gigthis (Médenine, 51 ha); Menzel Bourguiba Naval Academy (Bizerte, 65 ha); Base Militaire de Sidi Ahmed (Bizerte, 883 ha); Sufetula (Kasserine, 35 ha); Cillium (Kasserine, 30 ha); Thuburbo Majus (Zaghouan, 48 ha); Utica (Bizerte, 77 ha); Antonine baths (Tunis, 8 ha); Military area, Bizerte (Bizerte, 52 ha); Archaeological site, Zaghouan (Zaghouan, 13 ha); Dougga (Béja, 24 ha); Henchir el Beldia (Kairouan, 6 ha); قصر غرماسة (Tataouine, 25 ha) and 2 more.



# What OECMs could cover in KBAs

Tunisia has 72 KBAs; 70 are not fully covered by counted protected areas (6,174 km² uncovered). Priority 1 and 2 complexes reach **42** of them, cover **1,098 km²** of the gap and close at least half of it in **13**. Most of that is at sea (the trawling bans in Zembra, Thyna, Djerba and Kerkennah KBAs) and at dams.




The 22 KBAs with the largest uncovered area. Dashed line: 30%. Right: uncovered km² today.

## 23 KBAs reached by no candidate

They need protected areas or OECM types not mapped yet (forest and rangeland management, community conservation): Sebkhet Sidi El Hani (85 km²); Jbil (55 km²); Sebkhet Kelbia (32 km²); Garaet Mabtouh (21 km²); Sebkhet Sidi Khelifa (15 km²); Plaines de Kairouan (14 km²); Chaâmbi (11 km²); Sebkhet Ariana (11 km²); Salines de Monastir (4 km²); Lagune de Soliman (3 km²); Sebkhet Sejoumi (3 km²); Ichkeul (3 km²); Lac de Tunis (3 km²); Garaet Douza (2 km²); Sebkhet Nouaïel (2 km²); Lagunes de Korba (1 km²); Metbassta (1 km²); Réserve Naturelle Jebel El Ghorra (1 km²); Barrage Oued El Haajar (1 km²); Ghidma (1 km²); Douz Laâla (0 km²); Aqueduc de Zaghouan (0 km²); Barrage Sidi Abdelmonem (0 km²).

# What each kind of measure can offer

## Dams and reservoirs

338 records · 320 complexes · priority 1: 7 · priority 2: 25 · priority 3: 284

State-owned and permanent, managed by DGBGTH and the regional commissariats; several are KBAs for the White-headed Duck and other waterbirds. The Water Code (art. 123) bans polluting activities in 10 m and 10–60 m belts around drinking-water reservoirs, and some reservoirs are hunting reserves. **Recommendation:** pilot OECM recognition at the seven priority-1 dams with waterbird counts and an agreed management of the shores.

Their weakness is that they are managed for water, not biodiversity: water levels swing with irrigation demand, fishing is licensed, and grazing and disturbance on the shores are rarely managed. Recognition should come with a simple agreement on the shores and a commitment to count waterbirds every winter, which the International Waterbird Census already does at several sites.

## Fisheries closures

19 records · 19 complexes · priority 1: 3 · priority 2: 5 · priority 3: 4

The no-fishing zones around Zembra and La Galite are permanent and strict; most of their area is already inside the national park. The permanent trawling bans are very large and gear-specific. **Recommendation:** report the marine park zones; build the evidence on the trawling bans (enforcement, seagrass) before deciding.

The coral-harvesting bans and the seasonal closure of the Gulf of Gabès remove a single pressure or apply for a few months; they are kept as context. The no-fishing zones around offshore oil installations are small but strict and permanent.

## Hunting reserves

142 records · 142 complexes · priority 1: 1 · priority 2: 19 · priority 3: 122

Listed each season in the hunting order (art. 12). Named sites listed every season (dams, forests) can qualify; whole imadas include villages and farmland; rotating reserves are temporary. **Recommendation:** give stable reserves a permanent status and map their legal limits; narrow imadas to forest and rangeland.

Of the 142 mapped reserves, 72 are rotating or new (listed in a single season) and fail the long-term criterion; 64 whole imadas are listed every season but include villages and farmland. The strongest are the named sites: the wetlands of Sidi El Barrak and of the Lebna, Mlaabi and Abdel Monem dams, the islands of the Sidi Salem reservoir, the forest and catchment of the Siliana dam.

## Traditional oases

318 records · 64 complexes · priority 1: 0 · priority 2: 23 · priority 3: 41

Agricultural safeguard zones (Law 83-87) with a three-storey agro-ecosystem and old date varieties; Gafsa is a GIAHS site. **Recommendation:** document agro-biodiversity and water governance with the oasis water-user groups; start with Gafsa and the Nefzaoua oases in the Chott Djerid KBA.

The 318 palm-grove polygons were detected from satellite tree cover and grouped into 64 complexes; 23 reach priority 2, most in Kébili, where the Chott Djerid KBA outline includes the oasis fringe. Recognition would honour a living cultural landscape, but it rests on showing that traditional management maintains its diversity in the face of falling groundwater and urban spread.

## Sacred natural sites

1,066 records · 1,066 complexes · priority 1: 0 · priority 2: 14 · priority 3: 630

Trees around shrines are traditionally not cut, but the grove outlines are automatic detections and about one in three looked natural on imagery. Cemeteries are permanent but mostly without vegetation. **Recommendation:** verify the high-scoring groves; work with custodians and communities.

A few groves in the Mogods and Kroumirie score among the highest of all candidates because they are wooded and inside KBAs. Their outlines, however, are automatic detections around shrines: they are kept at priority 2 until checked.

## Archaeological sites

321 records · 321 complexes · priority 1: 1 · priority 2: 12 · priority 3: 62

Protected by the Heritage Code (Law 94-35) and managed by the INP; natural vegetation is often kept. **Recommendation:** add biodiversity monitoring to heritage management at sites with natural cover, starting with Henchir Bourgou.

Most mapped sites are small (a theatre, a temple, a

# What each kind of measure can offer (continued)

necropolis); 75 reach priority 1 to 3, among them Gigthis, Sufetula, Cillium, Thuburbo Majus, Utica and Dougga. Heritage management and biodiversity can reinforce each other where natural vegetation survives among the ruins.

## Military areas

18 records · 18 complexes · priority 1: 0 · priority 2: 4 · priority 3: 13

Restricted access over long periods, but many are bases or airfields; the southern border zone is a security measure renewed yearly. **Recommendation:** an informal discussion with the Ministry of Defence on the undeveloped ranges.

The 15 mapped military areas (over 50 ha) include air bases, barracks and a naval academy; four reach priority 2, among them the Sidi Ahmed base near Bizerte, which is mostly grassland. The southern border zone (2.5 million ha) is capped at priority 3.

## Cultural and historic sites

184 records · 184 complexes · priority 1: 0 · priority 2: 1 · priority 3: 25

Mostly built heritage with little habitat. **Recommendation:** context only.

Ksour, forts and medinas are important heritage but offer little habitat; only one reaches priority 2.

### KEY TAKEAWAYS

1. **Strongest**
   dam reservoirs in KBAs: state-owned, permanent, already partly protected by the Water Code
2. **Largest**
   the permanent trawling bans, if enforcement and seagrass outcomes can be shown
3. **Most numerous**
   sacred sites and cemeteries, but mostly too small or without natural vegetation
4. **Most delicate**
   military land and the border zone: consent of the Ministry of Defence
5. **Needs evidence**
   oases and hunting reserves: biodiversity and permanence to document

### OECM POTENTIAL AT A GLANCE

| | Strength | Perman. | Confid. | P1+P2 |
|---|---|---|---|---|
| Dams & reservoirs | 0.55 | 0.80 | 0.75 | 32 |
| Fisheries closures | 0.90 | 0.80 | 0.80 | 8 |
| Hunting reserves | 0.50 | 0.65 | 0.50 | 20 |
| Traditional oases | 0.40 | 0.80 | 0.60 | 23 |
| Sacred natural sites | 0.55 | 0.90 | 0.70 | 14 |
| Archaeological sites | 0.45 | 0.90 | 0.60 | 13 |
| Military areas | 0.55 | 0.80 | 0.55 | 4 |
| Cultural & historic sites | 0.25 | 0.90 | 0.60 | 1 |

Best measure type in each category (expert rubric, 0–1: strength, permanence, mapping confidence). Darker green: stronger. Last column: priority-1 and 2 complexes.

# A national OECM process, step by step

| When | What | Who |
|---|---|---|
| 0–6 months | **Quick wins and set-up.** Report Zembra zone C and the Kneiss islands to the WDPA. Create the OECM working group; agree the recognition procedure, the evidence expected and how OECMs are reported. | Ministry in charge of the environment, APAL |
| 6–18 months | **Pilots.** Assess the 9 priority-1 complexes with their authorities: boundaries, consent, waterbird and species data, management of the shores. Compile the evidence on the trawling bans. | DGBGTH, DGF, DGPA, INSTM, INP, CRDA; NGOs and universities |
| 18–36 months | **First recognitions and scale-up.** Recognise and report the pilots that meet the criteria; assess priority-2 complexes by category (oases, hunting reserves, sacred groves, archaeological sites). | Working group; CRDA, GDA, communities |
| To 2030 | **Close the gaps.** New protected areas and other OECM types for the KBAs no candidate reaches; yearly update of indicator 3.1. | Ministry in charge of the environment, DGF, partners |




| Who | Role |
|---|---|
| Ministry in charge of the environment (CBD focal point) | Leads the process, reports to the CBD and the WDPA/WD-OECM |
| APAL | Marine and coastal protected areas; reporting of marine park zones |
| DGF (Ministry of Agriculture) and regional forest districts | Hunting reserves, forests, national parks; waterbird counts |
| DGPA and INSTM | Fisheries closures, enforcement and habitat evidence at sea |
| DGBGTH and SONEDE | Dams, Water Code protection zones, drinking-water reservoirs |
| INP and AMVPPC | Archaeological and heritage sites |
| Ministry of Defence | Military areas and the border zone |
| CRDA, oasis water-user groups (GDA), communities | Oases, sacred groves, local governance and consent |
| NGOs and research (e.g. AAO/BirdLife Tunisia, universities), WWF | Biodiversity data, monitoring, facilitation |

## Making it work

**One procedure, many owners.** Each OECM stays under the authority that governs it today: the dam administration, the forest service, the fisheries administration, the heritage institute. The national working group's role is to agree a common procedure (what evidence, what boundary, what reporting) and to check each case against the CBD criteria, not to take over management.

**Start small and well documented.** A handful of strong, well-documented pilots will do more for the credibility of Tunisian OECMs than a long list reported at once. The priority-1 reservoirs are suited to this: the state owns them, the Water Code applies, waterbirds are counted, and the KBA status is established.

**Monitor what matters.** Outcomes should be followed with data the country already collects or can collect cheaply: winter waterbird counts on reservoirs, seagrass and fish surveys at sea, vegetation surveys on heritage sites, records of hunting and fishing offences. The same data feed the national reports to the CBD.

**Keep the data open and current.** Official boundaries, the legal texts that create each zone, and the yearly hunting orders are the backbone of this analysis; publishing them in usable form would let the screening be repeated each year, and extended to Algeria, Morocco, Libya and Egypt with the same tools.

ANNEX

# Method details and sources

| Measure type | Category | Strength | Permanence | Confidence | Cap |
|---|---|---|---|---|---|
| No-fishing zone | Fisheries closures | 0.90 | 0.80 | 0.80 | – |
| Permanent trawling ban | Fisheries closures | 0.60 | 0.80 | 0.60 | P2 |
| Seasonal trawling closure | Fisheries closures | 0.45 | 0.25 | 0.60 | – |
| Red-coral harvesting ban | Fisheries closures | 0.30 | 0.80 | 0.45 | – |
| Marine protected area zone not counted | Fisheries closures | 0.65 | 0.75 | 0.70 | – |
| Border buffer zone | Military areas | 0.45 | 0.50 | 0.40 | P3 |
| Military area | Military areas | 0.55 | 0.80 | 0.55 | – |
| Hunting reserve on a named site (every season) | Hunting reserves | 0.50 | 0.65 | 0.50 | – |
| Hunting reserve on a whole imada (every season) | Hunting reserves | 0.30 | 0.55 | 0.50 | P2 |
| Rotating or new hunting reserve | Hunting reserves | 0.30 | 0.15 | 0.50 | – |
| Water Code protection zone | Dams & reservoirs | 0.55 | 0.80 | 0.55 | – |
| Dam reservoir or hill lake | Dams & reservoirs | 0.30 | 0.80 | 0.75 | – |
| Traditional oasis | Traditional oases | 0.40 | 0.80 | 0.60 | – |
| Modern date-palm plantation | Traditional oases | 0.20 | 0.70 | 0.50 | – |
| Possible sacred grove | Sacred natural sites | 0.55 | 0.80 | 0.25 | – |
| Cemetery | Sacred natural sites | 0.35 | 0.90 | 0.70 | P2 |
| Archaeological site | Archaeological sites | 0.45 | 0.90 | 0.60 | – |
| Cultural or historic site | Cultural & historic sites | 0.25 | 0.90 | 0.60 | P2 |

## Scoring

Score = 0.45 × Measure + 0.40 × Biodiversity + 0.15 × Size + 0.03 per extra stacked category (max 2). Measure = 0.45 strength + 0.35 permanence + 0.20 confidence. Biodiversity = 0.5 KBA + 0.3 habitat + 0.1 representation + 0.1 connectivity; the KBA term is the share of the net area in KBAs (full at 50%), multiplied by the habitat term. Size = log10(1 + ha) / log10(10,001).

## Sources

OECM Harvester outputs for Tunisia (OpenStreetMap, Wikidata, GeoNames, Global Dam Watch, ESA WorldCover, GFCM, ProtectedSeas, national legal texts) · WDPA/WD-OECM, October 2026 (UNEP-WCMC & IUCN) · World Database of KBAs, September 2026 (© KBA Partnership) · ESA WorldCover 2021 · RESOLVE Ecoregions 2017, MEOW · geoBoundaries · CBD Decision 14/8 · IUCN-WCPA (2019) Recognising and reporting OECMs.

## Files

tunisia/06_oecm: scripts (oecm_measures.yaml holds every judgement), outputs (oecm_complexes.csv / .gpkg, oecm_kbas.csv) and the interactive dashboard (north_africa_dashboard/wwf_site_oecm, tab OECM opportunities).

### GLOSSARY

**OECM**
Other effective area-based conservation measure: a place outside protected areas that achieves lasting in-situ conservation (CBD Decision 14/8).

**KBA**
Key Biodiversity Area: a site contributing significantly to the global persistence of biodiversity, identified with the KBA Standard.

**Indicator 3.1**
GBF Headline Indicator: share of land and sea covered by protected areas and OECMs.

**WDPA / WD-OECM**
World Databases on Protected Areas and on OECMs (UNEP-WCMC, IUCN), the source of indicator 3.1.

**Net area**
The part of a site outside the protected areas already counted: what it would add to 3.1.

**Complex**
Candidate records of the same place grouped and assessed together (e.g. a dam, its Water Code belts, a hunting reserve on it).

**Trigger species**
A species for which a KBA was identified. CR, EN, VU: Critically Endangered, Endangered, Vulnerable (IUCN Red List).

**Reporting gap**
A protected area that exists but is missing from the WDPA records counted in 3.1.